High-Resolution Detection of Stratospheric Aerosols in CALIPSO Atmospheric Lidar Data Facilitated by the CALIOP-Density-Dimension Algorithm

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***Abstract*— The objective of this article is to demonstrate a new capability to detect faint stratospheric aerosols in atmospheric Lidar data from NASA's Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observations (CALIPSO) Mission (June 12, 2006 to June 30, 2023) at high resolution. Faint aerosols from wildfires, distant volcanic eruptions, and Asian and Saharan dust storms play important roles in the Earth's atmosphere system, as they provide essential air quality and pollution, airline safety, climate radiative forcing, monitoring of fires and volcanic eruptions, environmental safety, and weather forecasting. Tenuous aerosols are recorded in the CALIOP atmospheric Lidar data from the CALIPSO mission, but especially extremely faint stratospheric aerosol layers often escape detection and classification partly or entirely in the current CALIOP data analysis scheme. To solve this problem, we introduce a new algorithm for detection and height determination of atmospheric layers, the CALIOP-Density-Dimension Algorithm (CALIOP-DDA). The DDA is an autoadaptive algorithm that builds on concepts from artificial intelligence and spatial statistics. Core steps are calculation of a density field and application of a threshold function for signal–noise separation. The stratospheric aerosol detection is aided by the tropopause split concept (TSC). The CALIOP-DDA facilitates detection of extremely faint stratospheric aerosols from various sources, including distant wildfires and volcanic eruptions, in nighttime and daytime CALIOP data, even in the presence of complex types of other cloud and aerosol layers across a large range of optical thicknesses, while retaining the full 334-m along-track and 30-m height resolution, without creating false positives. CALIOP-DDA results are evaluated by comparison with layer heights derived from airborne validation experiments, conducted using the cloud-physics Lidar (CPL). In conclusion, the CALIOP-DDA holds promise as the algorithmic basis for a future improved, high-resolution CALIPSO data product.**

***Index Terms*— Aerosol detection, atmospheric Lidar, Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observations (CALIPSO), density-dimension algorithm, signal processing, smoke plumes, volcanic dust, wildfires.**

I. Introduction

A. Importance of Detection and Height Measurement of Tenuous Aerosols

FAINT aerosols from wildfires, distant volcanic eruptions, and Asian and Saharan dust storms play important roles in the Earth's atmosphere system, as they provide essential air quality and pollution, airline safety, climate radiative forcing, monitoring of fires and volcanic eruptions, environmental safety, and weather forecasting [1]. Radiative forcing from aerosols remains the largest uncertainty in climate forcing overall, as established by the Intergovernmental Panel on Climate Change (IPCC) in its sixth Assessment Report [2], [3], [4]. Detection of altitude and extent of aerosol layers in the atmosphere provides the basis for estimates of aerosol optical depth (AOD), a quantity widely used as a constraint on climate models or for model evaluation (see [5], [6], [7], [8], [9]). However, significant uncertainties exist in observed AOD determinations derived from different sensors [10], [11], [12], [13], which in turn leads to uncertainties in the climate models that rely on these estimates, a problem that highlights the need for continued efforts to improve satellite estimates of aerosol layers and optical depth [13], [14], [15], [16].

B. Measurements of Atmospheric Aerosols and the Problem of Detection of Tenuous Layers

Aerosol layers in the atmosphere can be measured by ground-based sensors, in situ airborne sampling, passive satellite imagers, or, most relevant to this work, active satellite sensors. Each of these approaches has distinct strengths and weaknesses that must be understood in order to determine their usefulness and potential impacts. The most obvious benefit of observations from a polar-orbiting satellite is the global

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coverage of the retrievals, and the ability to resolve them spatially. Examples of active remote-sensing satellite missions whose data have been widely used include the following.

1) NASA's CloudSat (2006–2023) mission, with the cloud profiling radar (CPR) as the main instrument, a 94-GHz nadir-looking radar that measures the power backscattered by clouds as a function of distance from the radar.
2) The Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observations (CALIPSO) mission [17], [18] (2006–2023), which carried the Cloud-Aerosol Lidar with Orthogonal Polarization (CALIOP), a two-wavelength (532 and 1064 nm), polarization-sensitive (at 532 nm) backscatter Lidar.
3) The current NASA Ice, Cloud, and Land Elevation Satellite-2 (ICESat-2) Mission, which operates the Advanced Topographic Laser Altimeter System (ATLAS), a 532-nm multibeam micropulse photon-counting laser (Lidar) altimeter [19], [20].

CALIPSO data have been used to understand the climatology of aerosol and cloud layers around the world—including their vertical distribution, which cannot be determined from passive sensor data, such as Moderate Resolution Imaging Spectrometer (MODIS) data [21]. CALIPSO collects data during daytime and nighttime [13].

Research using CALIPSO data has been very successful in a wide range of atmospheric and climatic science topics, building on its measurement properties. Because of its global domain and vertical resolution, CALIPSO data can be used to contextualize and extrapolate other ways of understanding clouds and aerosols. This includes providing checks on global climate models to ensure that their predicted aerosol and cloud layer concentrations and forced parameters match with climatology; correlating in situ observations of physical, optical, and chemical properties of aerosols to extrapolate from the campaign domain to global scales; and providing context and background values for local experiments [4], [22].

Aerosols and cirrus clouds are generally well-detected in the CALIOP data [23], [24], [25]. A classification scheme allows to distinguish different types of clouds and aerosols in close proximity or in complex settings, for example, aerosol layers with embedded clouds [26], [27]. But tenuous atmospheric layers with a low gradient and layers with poor signal-to-noise ratios are often missed by the current detection algorithm [13], [28]. Balmes and Fu [28] and Davis et al. [29] established an operational missing optical depth threshold of 0.01 for CALIPSO data (for comparison, MODIS begins to experience a significant drop-off in detection sensitivity at cloud optical depths of $\approx$0.4 [30], and the threshold for subvisible clouds is 0.03 [31]). Such layers are typically very diffuse aerosols in the lower stratosphere and in the troposphere and volcanic dust and subvisible cirrus clouds in the upper troposphere, whose detection is important for a large range of societal and science problems as highlighted in Section I-A. Though even optically thin aerosol layers, as recorded in the CALIOP data, can be picked out in CALIPSO data scenes by eye, algorithmically separating such small signals from the noise is a difficult problem. The lack of detection results in a roughly 13% lower estimate of AOD than is found with other instruments [13]. CALIPSO column AOD retrievals have been found to be lower than those obtained from MODIS [10], [12], [32]. This can be due to a number of factors, including error in assigned extinction to backscatter ratio, cloud contamination, and inability to detect very thin layers [32]. Considering that CALIOP only retrieves extinction coefficients in those parts of a profile, where the signal-to-noise ratio is sufficiently high to reliably detect a feature, Toth et al. [33] revisited the minimum aerosol layer detection sensitivities and analyzes their subsequent impacts on aerosol optical thickness retrievals in CALIPSO Level-2 data products, which resolves part of the difference in AOD retrievals between CALIPSO and MODIS data. While all of these factors may contribute to this discrepancy, the impact of undetected aerosol layers is both significant and underappreciated [34]. Resolving this requires the development of a layer detection algorithm capable of identifying these layers, which is the primary aim of this work.

The current family of CALIOP data processing algorithms is built upon a core of algorithms that defined state-of-the-art for Lidar retrieval techniques at the time of launch, and while it has been continuously upgraded, the core algorithms have remained unchanged [35]. The need for an advanced, modern algorithm that can facilitate the detection of hitherto missed tenuous layers motivates the work in this article (Section I-C). Since the first release of the CALIOP data products, several alternative approaches have been proposed (see [36], [37], [38], [39]). While these methods have all been evaluated in terms of accuracy, multilayer sensitivity, implementation complexity, computational speed, and compatibility with CALIOP existing feature classification algorithms, so far, no replacement algorithm has been identified.

C. Objectives of This Study

The objective of this article is to demonstrate a new capability to detect faint stratospheric aerosols in atmospheric Lidar data from NASA's CALIPSO Mission (April 28, 2006–August 1, 2023), afforded by the development of a new algorithm, the CALIOP-Density-Dimension Algorithm (CALIOP-DDA). The CALIOP-DDA utilizes CALIOP level L1B data total attenuated backscatter (TAB) data at 532 nm (532T) as input into its analysis, signal classification and layer detection scheme and results in: 1) a signal–background separation (classification of signals); 2) detection of atmospheric layers; 3) height measurements of atmospheric layers in the form of layer boundary heights (top and bottom); and 4) a confidence measure. The CALIOP-DDA also provides density fields, whose role in the detection scheme will be described in Section II. The analysis in this study focuses on the optimization and application of the CALIOP-DDA for the detection of faint stratospheric aerosols. DDA analysis results will be evaluated using airborne data collected with the cloud physics Lidar (CPL) during the 2009 CALIPSO validation campaign.

CALIPSO collected atmospheric Lidar data in three channels (1064, 532, and 350 nm), with the 532nm observations

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including a parallel and a perpendicular (normalized) polarization. Data were downlinked as 532T (532nm total attenuated backscatter, sum of 532nm parallel and perpendicular), 532S (532nm perpendicular), 1064T (1064nm total attenuated backscatter) and 350nm (not used much). The current CALIOP layer detection algorithm uses only the 532T attenuated backscatter profile in its search for clouds and aerosols in the Earth's atmosphere [35], [40]. Thus, we will base the detection studies in this article also on 532T, to focus on the development and evaluation of a new algorithm for high-resolution detection of faint stratospheric aerosols. Results from the CALIOP-DDA layer detection will be compared with results from the current CALIOP layer detection algorithm.

II. Motivation and Summary of the Approach for High-Resolution Detection and Height Measurement of Faint Stratospheric Aerosols

The motivation and approach for the work in this article is illustrated by the situation of a complex scene of CALIPSO data (see Fig. 1), where a faint stratospheric smoke plume is recorded in the low-level CALIOP data at 17–20-km altitude [see Fig. 1(a)]. The signal of the smoke plume in the L1B CALIOP TAB data is 2.5 orders of magnitude lower than that of cloud layers in the same scene. In addition, the backscatter values of the background (noise) at 2–10-km altitude are of the same or higher level of intensity as those of the stratospheric smoke plume, increasing with distance from the sensor, i.e., increasing with decreasing altitude. This situation prevents a detection based on absolute backscatter intensity. The low gradient of the smoke plume relative to backscatter values recorded in its neighborhood indicates a mathematically difficult detection problem. The current detection and classification capability, resultant from the CALIOP algorithms and reported on the CALIOP product as the vertical feature mask (VFM), is illustrated in Fig. 1(d). The stratospheric smoke plume is partially detected, but largely missed and often misclassified (as cloud). The CALIOP detection scheme utilizes a data aggregation based on height and intensity-dependent averaging of recorded signals (Section III-D), which leads to the coarse, 80-km resolution representation of high-altitude layers in the VFM in this example.

To solve the problem of detection and height measurement of tenuous atmospheric layers, we develop the Density-Dimension Algorithm for CALIOP atmospheric Lidar data from NASA's CALIPSO Mission, the CALIOP-DDA. The CALIOP-DDA is based on the Density-Dimension Algorithm for atmospheric data collected with the ATLAS of NASA's Ice, Cloud, and Land Elevation Satellite-2 (ICESat-2) mission, termed DDA-atmos for short and described in [41] and in detail in [42]. The DDA builds on concepts from artificial intelligence and spatial statistics. Core steps are the calculation of a density field and the application of a threshold function for signal–noise separation. The DDA is an autoadaptive algorithm, in that it automatically adapts to changing background conditions of day times and reflectance. The CALIOP-DDA is introduced in this article in a summative fashion (see Section IV-C). The idea of the DDA

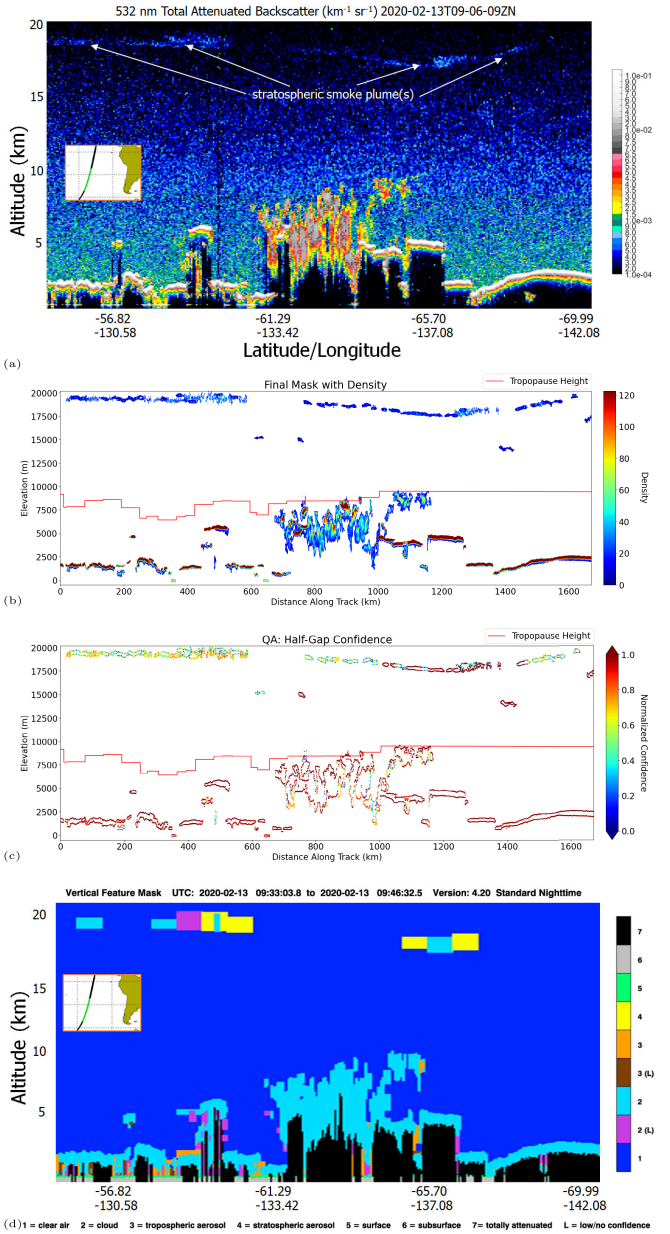


Fig. 1. Demonstration of detection of faint stratospheric smoke plumes in CALIOP data v4.51 using the CALIOP-DDA. The smoke plume is detected and reported at dataset resolution, although its signal in the TAB data is 2.5 orders of magnitude lower than that of cloud layers in the same data records. CALIOP dataset from Feburary 13, 2020. Subset of granule CAL_LID_L1-Standard-V4-51.2020-02-13T09-06-09ZN.h5. Nighttime Data. (a) 532-nm TAB data. Inset: Data location. (b) Result of autoadaptive layer detection using CALIOP-DDA. The dark-blue layer on top is the stratospheric smoke plume. The green ranges—thin clouds. Red/orange ranges: optically thick clouds. Resolution of detected layers is 335-m along-track/30-m height (see text). (c) Half-gap confidence of cloud and aerosol layer detection. (d) Current detection and classification capability given in VFM of the CALIOP data product. Note along-track resolution is 80-km along-track. Note the stratospheric smoke plume is partially detected, but largely missed and often misclassified (VFM v4.20 used for display purposes).

is the notion that a signal point is likely surrounded by other signal points, thus forming a center of densely recorded Lidar returns, whereas background regions are characterized by randomly distributed high and low values. Thus, a density field is calculated, using each point as a density center,

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TABLE I
RESOLUTION OF CALIOP DATA

altitude range	top index	base index	shots averaged (along track distance)	bins averaged (vertical distance)
39.8 – 30.2	1	33	15 (5 km)	20 (300 m)
30.0 – 20.3	34	88	5 (5/3 km)	12 (180 m)
20.2 – 8.2	89	288	3 (1 km)	4 (60 m)
8.2 – -0.5	289	578	1 (1/3 km)	2 (30 m)
-0.6 – -1.8	579	583	1 (1/3 km)	20 (300 m)

and a threshold function can be utilized to separate signal from background. Ultimately, this approach to data aggregation allows the retention of the high resolution of the originally recorded, low-level Lidar data. The DDA-atmos has proven efficient for the detection of tenuous clouds, blowing snow, and diamond dust in ICESat-2 atmospheric data [41], [43]. The DDA-atmos was selected as the official operational algorithm for atmospheric layer detection in ICESat-2 data, because it excels at detecting the tenuous layers, in addition to and in the presence of optically thick layers. Results are reported on ICESat-2 ATLAS data product ATL09 [44].

Other than the atmospheric data from ICESat-2, which only extends to 14-km altitude above the Earth's surface, CALIPSO records data to almost 40 km, but recordings of returns from high-altitude features are increasingly affected by changing signal-to-background characteristics with increasing altitude. Total attenuated backscatter becomes increasingly noisy with distance from the sensor (with increasing range, i.e., decreasing altitude), whereas attenuated scattering ratio (ASR, as defined in Section IV-B) is decreasingly affected by range. This situation creates a unique detection challenge for the problem of retrieval of stratospheric aerosols and tenuous, high cirrus clouds in CALIPSO data, which is further acerbated by the difference between daytime and nighttime data. CALIOP data have a better signal-to-background ratio for daytime data than ICESat-2 atmospheric data, but worse signal-to-background ratio for nighttime data, while of course, background is much higher for daytime data in general than for nighttime data. To solve the detection problem especially for difficult cases, we introduce a physically based concept, termed "tropopause split" concept. Tropopause height, as reported on the CALIOP L1B data, is utilized to split the dataset and apply the DDA in two altitude regions, then combine the results, taking care to avoid edge effects around the tropopause height (Section IV-D).

Application of the CALIOP-DDA with tropopause split to the smoke plume dataset leads to detection of the faint stratospheric smoke plume, even in locations where it is not detectable by eye in the input data [see Fig. 1(b)]. The resolution of the smoke plume's spatial characteristics is better than that of the data recorded at this altitude (see Table I and Section IV-D) and the same as the 334-m along-track/ 30-m height resolution of the L1B data in the 0–8.2-km altitude range. The smoke plume is detected with high (0.8–1) to medium (0.5–0.7) confidence [see Fig. 1(c)]. This summary of the approach is intended to serve as a motivation for the more technical sections of this article.

III. CALIPSO MISSION AND RESULTANT DATA

A. Mission and Instrumentation

The CALIPSO mission was a unique investigation of the vertical structure of the atmosphere conducted jointly by the national space agencies of U.S. (NASA: National Aeronautics and Space Administration) and France (CNES: Centre National d'Etudes Spatiales). CALIPSO was operational from April 28, 2006 to August 1, 2023 (science data available June 12, 2006 to June 30, 2023) with four primary science goals (see http://www-calipso.larc.nasa.gov/about/science.php, [17], [18]).

1) Providing observationally based estimates of direct aerosol radiative forcing and quantifying the uncertainties in these estimates.
2) Providing an empirical basis for assessing indirect aerosol radiative forcing and (again) quantifying the uncertainties in these estimates.
3) Improving the accuracy of satellite-based estimates of long-wave radiative fluxes obtained from passive sensors.
4) Obtaining the first-ever global, height-resolved measurements of thin cirrus, polar clouds, and multilayered cloud systems, thus enabling improved assessments of cloud-climate feedbacks.

To accomplish these goals, CALIPSO augmented the existing A-Train satellite measurements [45] with three new instruments: a single channel (645 nm) wide field-of-view camera, the three-channel (8.65, 10.6, and 12.0 μm) imaging infrared radiometer (IIR), and the CALIOP. CALIOP was a two-wavelength (532 and 1064 nm), polarization-sensitive (at 532 nm) elastic backscatter Lidar that makes range-resolved measurements of clouds and aerosols in the Earth's atmosphere from 40 km above mean sea level down to the planetary surface. CALIOP has provided the high-resolution vertical profile measurements that were essential for achieving the mission science goals. Among the most critically important geophysical parameters derived from these measurements are the vertical locations (i.e., top and base altitudes) of cloud and aerosol layers. These data are fundamentally required by all downstream algorithms in the CALIOP retrieval scheme, are an essential component of the IIR science retrievals, and inform every aspect of the CALIPSO science goals.

B. CALIOP L1B Lidar Data: Attenuated Backscatter

The CALIOP Lidar transmits polarized light at 532- (green) and 1064-nm [infrared (IR)] wavelengths and has three receive channels for the attenuated backscattered signal from molecules and particles in the atmosphere. Both the parallel and perpendicular components of the 532-nm received signal, along with the total 1064-nm signal, are processed, geolocated, and calibrated to provide three Lidar profiles of attenuated backscatter as the primary datasets of the Level 1B (L1B) data product [46], [47], [48]. Attenuated backscatter profiles are reported for each laser pulse as an array of 583 elements, registered to a constant altitude grid (see Table I), with units of $1/(\text{km} \cdot \text{sr})$ (sr = steradians). Specifically, the

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three attenuated backscatter profiles are: 1) TAB for 532-nm light, 532T ($\beta_{532T} = \beta_{532\parallel} + \beta_{532\perp}$, both parallel and perpendicular polarization components) and 2) perpendicular attenuated backscatter for 532-nm light, 532S ($\beta_{532\perp}$, returned signal polarized perpendicularly to transmit signal polarization), and attenuated backscatter for 1064-nm light, 1064T (β_{1064}, no polarization component in measurement).

A single L1B data product contains calibrated and geolocated Lidar measurements from half of an orbit, containing either full day or night segments for the three channels. The can be applied CALIOP-DDA to any of the three attenuated backscatter estimates within the CALIPSO Lidar L1B profile data as input to the algorithm. All data processing to this point is carried out as part of the CALIOP standard data processing. Our analysis uses Version 4-51 (v4-51) of the L1B data [49] and uses only the β_{532T} profiles. An example of the resultant TAB data for 532-nm light, β_{532T}, is shown in Fig. 1(a).

C. CALIOP Data Height-Dependent Resolution

CALIOP data are recorded with an along-track and height resolution that depends on five altitude ranges (see Table I). In this article, we analyze data from the [−0.5, 8.2] and the [8.2–20.2] km altitude ranges (relative to the WGS84 ellipsoid, see Geolocation Assessment Algorithm for CALIPSO Using Coastline Detection [50]).

D. Operational CALIOP Layer Detection Scheme

The core of the current CALIOP layer detection scheme [35] is an extensively validated profile scanning engine that uses a built-in awareness of the physics of light scattering in a turbid atmosphere to automatically adapt to changing noise environments and the attenuation effects inherent in multilayered scenes. This profile scanning engine is driven by a multiresolution spatial averaging scheme that identifies increasingly fainter features at increasing coarser resolutions. The outputs of this algorithm provide the inputs for successive analyses that first discriminate between cloud and aerosol layers [26], then ascertain cloud thermodynamic phase [51] or classify aerosols by type [12], and ultimately retrieved profiles of particulate extinction coefficients and estimates of layer optical depths [52]. The results obtained from this processing chain are reported in the CALIPSO Level-2 data products [53] and to date have been used in over 4000 peer-reviewed journals to investigate an extensive range of scientific questions that address numerous aspects of the mission science goals.

The CALIOP feature detection scheme uses a threshold-based algorithm that averages the backscatter signal to multiple horizontal resolutions ranging from 1/3 up to 80 km depending on the signal strength of the feature [35]. Optically thick clouds can be detected at the 1/3-km resolution of the data below 8.2 km. Aerosols, typically much more faint than clouds, require more averaging for detection, resulting in horizontal resolutions of 20 or 80 km in most cases. After a layer is detected, the cloud-aerosol discrimination (CAD) algorithm classifies each layer as either cloud or aerosol based on its geolocation and optical properties [26]. These aerosol layers are classified further as either tropospheric or stratospheric based on their altitudes with respect to the tropopause height. These basic classifications, along with a quality assessment (QA) for each layer, are reported on the CALIPSO Level-2 VFM, v4-51 data product [54]. The VFM reports layer detection and classification status for each range bin at the full spatial resolution of the downlinked data [55]. Features are further subtyped (e.g., ice/water phase and/or aerosol type [51], [56]), but for this analysis, we use the VFM results for comparisons to the detection capability and resolution of the current algorithm.

IV. CALIOP-DDA

A. Input Data

As input to the CALIOP-DDA, we utilize two types of CALIPSO data: 1) total attenuated backscatter data for 532-nm light, β_{532T}, abbreviated here as TAB, as briefly described in Section III-B and reported in the CALIOP L1B data product [49] and/or 2) attenuated scattering ratio (ASR) data, which are derived from TAB data as described in Section IV-B.

The CALIPSO data products also include, among a lot of other types of information, a classification scheme with the following classes: 1) clear air; 2) cloud; 3) tropospheric aerosol; 4) stratospheric aerosol; 5) surface; 6) subsurface; and 7) totally attenuated, where classes 2) and 3) have additional subclasses of low confidence (2L, class 0) and (3L, class-1) [see Fig. 1(d)]. The detection and classification results, represented in the VFM (see Section III-D) will be utilized in this article as the baseline for comparison between the detection capabilities of CALIOP-DDA and the CALIOP detection and classification scheme in the current version, v4.51 [54].

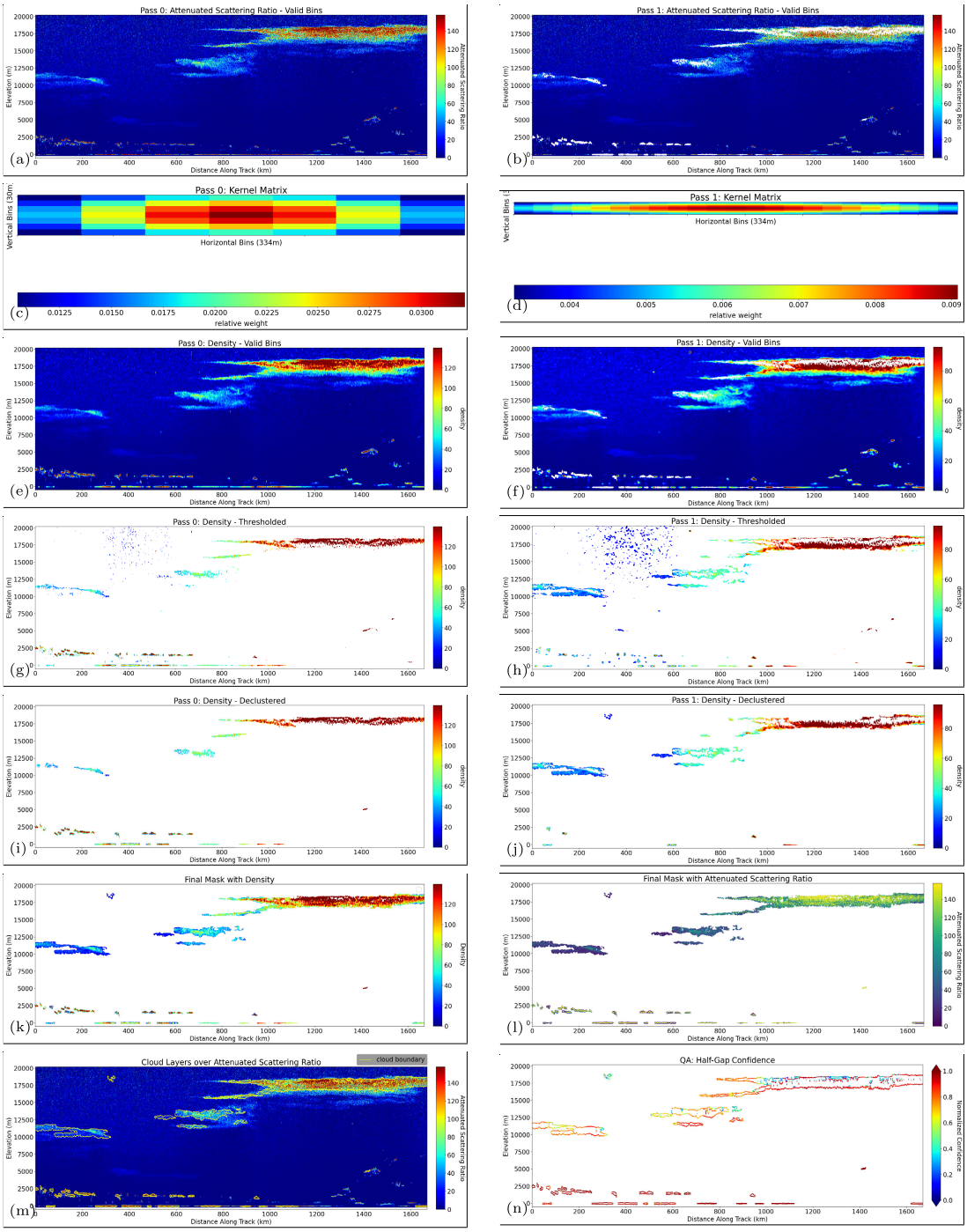
B. Optional Transformation to Attenuated Scattering Ratio (ASR)—The Trafo Function

Alternative to using CALIOP L1B TAB data as input to the DDA, experiments were carried out to investigate faint layer detection from ASR data. In the current algorithm version, we are using a CALIOP-data specific transformation scheme for transformation of TAB data to attenuated backscatter ratio based on [35].

Controlled by an input parameter, the CALIOP-DDA is given the option to convert the attenuated backscatter estimates to a so-called “attenuated scattering ratio" by dividing by a “clear-air” backscatter model as formalized in [35, eq. (3)]. The molecular backscatter model provides an idealized attenuated backscatter profile that is devoid of any clouds or aerosols, consisting only of molecular (Rayleigh) scattering with Rayleigh extinction and ozone absorption taken into account. Thus, in the absence of noise and in clear skies without any clouds or aerosols, the attenuated scattering ratios will have a range-independent value of 1. Values in excess of 1 are “features” (i.e., cloud or aerosol). This uniformity makes it theoretically easier to implement a search for layers in some arbitrary region of the atmosphere.

The molecular backscatter model uses additional data reported on the CALIOP L1B data product, namely, the molecular and ozone number densities ($\rho_m \in \mathcal{R}^n$ and $\rho_{O3} \in \mathcal{R}^n$, respectively), where n is the number of height bins (or range

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Run Name	s	a	u	t	T	m	L	q	Q	TF
base2	3,3	10,40	1	0.9, 1.0	0.0001	300,600	2	0.95,0.9	0.98,0.95	True

Fig. 2. 2009-02-10 Australian Fire Example, DDA-steps. CALIOP trafo function applied to CALIOP Data before DDA. Data in upper, low-resolution layer simply duplicated. This figure highlights the DDA steps for two density runs. Subset of CAL_LID_L1-Standard-V4-51.2009-02-10T12-33-03ZN.h5 (subset_profs_40000_45000_CAL_LID_L1-Standard-V4-51.2009-02-10T12-33-03ZN.h5). (a) Total attenuated backscatter (TAB) with the transfer function applied, resulting in attenuated scattering ratio (ASR). (b) ASR data with regions from pass 0 masked. (c, d) Kernel of pass 0 (density run 1) and pass 1 (density run 2). (e, f) Density field, pass 0 and pass 1, respectively. (g, h) Density fields 1 and 2, respectively, thresholded (after application of auto-adaptive thresholding function). (i, j) Density fields 1 and 2, respectively, thresholded and declustered. Density mask 1 and 2 (resp.). (k) Final mask with density. (l) Final mask with ASR. (m) Cloud layer boundaries over ASR data. (n) Half-gap confidence. CALIOP-DDA parameter table at bottom (s—standard deviation, a—anisotropy, u—cutoff, t—threshold factor, T—threshold bias offset, m—minimum cluster size, L—window size parameter, q—threshold quantile for daytime, Q—threshold quantile for nighttime; first value for each parameter corresponds to the first density run, second value for each parameter corresponds to the second density run).

bins) in a CALIOP data profile, the wavelength-dependent (λ) Rayleigh backscatter cross section ($\xi_{\beta,\lambda} \in \mathcal{R}$), the wavelength-dependent Rayleigh extinction cross section ($\xi_{\sigma,\lambda} \in \mathcal{R}$), and the ozone absorption cross section ($\xi_{O3} \in \mathcal{R}$).

The values of the height-dependent density vectors $\rho_m = (\rho_{m_i})_{i=1}^{n}$ and $\rho_{O3} = (\rho_{O3_i})_{i=1}^{n}$ are ancillary data stemming from NASA's Global Modeling and Assimilation Office (GMAO) [57], whereas the cross-sectional values are

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typically derived in simulations and do not depend on height (or range).

The modeled molecular attenuated backscatter values ($\beta_M = (\beta_{M_i})_{i=1}^n \in \mathcal{R}^n$) are estimated as the product of the altitude-dependent backscatter coefficient ($\beta_m = (\beta_{m_i})_{i=1}^n \in \mathcal{R}^n$) and the scalar product of the altitude-dependent molecular two-way transmittance ($T_m = (T_{m_i})_{i=1}^n \in \mathcal{R}^n$) with the altitude-dependent ozone two-way transmittance ($T_{O3} = (T_{O3_i})_{i=1}^n \in \mathcal{R}^n$).

That is,

$$\beta_{M_i} = \beta_{m_i} \cdot T_{m_i} \cdot T_{O3_i} \tag{1}$$

where β_{m_i} is given by

$$\beta_{m_i} = \xi_{\beta,\lambda} \cdot \rho_{m_i}. \tag{2}$$

The values of the height-dependent two-way transmittances T_m and T_{O3} are derived from the range-dependent two-way transmittances $T_m(r)$ and $T_{O3}(r)$, which are given by

$$T_m(r) = exp\left(-2 \cdot \int_0^r \xi_{\sigma,\lambda} \cdot \rho_m(r')dr'\right) \tag{3}$$

for the range-dependent molecular two-way transmittance $T_m(r)$ and by

$$T_{O3}(r) = exp\left(-2 \cdot \int_0^r \xi_{O3} \cdot \rho_{O3}(r')dr'\right) \tag{4}$$

for the range-dependent ozone two-way transmittance $[T_{O3}(r)]$. Thus, the attenuated scattering ratio for a given altitude bin (β_{ASR_i}) is determined as the ratio of the associated total attenuated backscatter value (β_{TAB_i}) reported in the CALIOP L1B product and the modeled molecular backscatter value (β_{M_i}), where

$$\beta_{ASR_i} = \frac{\beta_{TAB_i}}{\beta_{M_i}} = \frac{\beta_i}{\beta_{M_i}} \tag{5}$$

relating the ASR vector ($\beta_{\text{ASR}} = (\beta_{\text{ASR}})_{i=1}^n \in \mathcal{R}^n$) to the TAB vector ($\beta_{\text{TAB}} = (\beta_{TAB_i})_{i=1}^n \in \mathcal{R}^n$). In this article, we use TAB data from the 532T CALIPSO channel as $\beta_{\text{TAB}} = \beta_{532T}$, as introduced in Section IV-A.

C. Mathematical Principles of the CALIOP-DDA

1) Algorithm Steps: The DDA steps are illustrated in an analysis of the "2009 Australian Fire Golden Granule" (see Fig. 2). The so-called golden granules are a collection of test datasets for algorithm evaluation selected by Mark Vaughan and thus provide bench-mark results.

The core mathematical principles of the CALIOP-DDA follow those of the DDA-atmos for ICESat-2 data [41], with several adaptations necessitated by the unique data structure of the CALIOP data (see Sections III-B and III-C). Additional steps and modules developed for the CALIOP data processing and summarized in Sections IV-B and IV-E.

The core CALIOP-DDA algorithm proceeds by the following main steps. The result of each step is output as a data mask, and a mask handler combines those into a final mask, on which the cloud/aerosol layer boundary determination is based. The description here follows that in [41] with modification as needed for CALIOP data analysis. The main steps of the CALIOP-DDA are illustrated in an analysis of the "2009 Australian Fire Golden Granule" (see Fig. 2). The so-called golden granules are a collection of test datasets for algorithm evaluation selected by Mark Vaughan and thus provide benchmark results.

1) *Step 1:* Set algorithm-specific parameters and load data.
2) *Step 2:* Calculation of density field.
3) *Step 3:* Using density as a dimension: application of the autoadaptive threshold algorithm to separate atmospheric layers from background (noise).
4) *Step 4:* Removal of small clusters: determination of cloud and aerosol areas (cloud mask).

Step 1 (Set Algorithm-Specific Parameters and Load Data): The DDA is driven by a set of algorithm-specific parameters. Of importance for this article are the parameters of the kernel of the radial basis function (*rbf*) in the density calculation (see step 2) and of the autoadaptive threshold function (see step 3). For CALIOP data analysis, data loading is preceded by a number of preprocessing steps.

Step 2 (Calculation of Density Field): As described in the summary of approach (Section II), the main idea is that points within an atmospheric layer can be expected to have a higher density than background areas (regions of ambient light or noise outside of layers). Thus, identification of points within a layer requires data aggregation. This data aggregation is performed by application of a Gaussian *rbf*. A *rbf* is a real-valued function whose value decreases with distance from the center c see (1)

$$\Phi(x, c) = \Phi(\|x-c\|) \tag{6}$$

for all x in a definition area $\mathcal{D}$ with respect to any norm $\|\cdot\|$. The density field is calculated by letting each point in the atmospheric dataset take the role of a center, evaluating the *rbf* for all neighboring points, and forming the sum of weights from this process. In the algorithm, we utilize a Gaussian *rbf* (letting $r = x - c$ and $s \in \mathcal{R}$ for a radius r and standard deviation s)

$$\Phi(r) = e^{-\left(\frac{r}{\sqrt{2}s}\right)^2}. \tag{7}$$

The Gaussian *rbf* gives higher weights to close-by points and weights decrease with distance from the center point. For CALIPSO data, which are recorded in bins of dimensions as given in Table I (e.g., 334-m along-track and 30-m height), the density calculation is carried out for each bin in place of a center point. A weight matrix is calculated as follows:

$$W(c, x) = W_c(x) = \Phi(\|x-c\|_a) \tag{8}$$

with $x \in \mathcal{D}_c = \{\tilde{x} \in \mathcal{D}: \|\tilde{x} - c\|_\infty \leq r\}$. The set of all bins within a given rectangular box around the center bin c (in this initial distance determination, simply the infinity-norm (absolute distance in each direction) $\|\cdot\|_\infty$ is used). The kernel is illustrated in Fig. 2(c) and (d). Then the density value, $f_d(c)$, is calculated as follows:

$$f_d(c) = \sum_{x \in \mathcal{D}_c} W_c(x) z(x) \tag{9}$$

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where $z(x)$ is the bin data (input data, TAB or backscatter ratio). The normalized density value $f_d^{\text{norm}}(c)$ is

$$f_d^{\text{norm}}(c) = \frac{\sum_{x\in\mathcal{D}_c} W_c(x)z(x)}{\sum_{x\in\mathcal{D}_c} W_c(x)} \tag{10}$$

where the denominator works as a normalization factor. For atmospheric data analysis, we use a norm $\|\cdot\|_a$ that takes the anisotropy of most atmospheric layers into account, which have larger horizontal than vertical extension

$$\|x-c\|_a = \|A(x-c)\|_2 \tag{11}$$

with an anisotropy matrix

$$A = \begin{pmatrix} \frac{1}{a} & 0 \\ 0 & 1 \end{pmatrix} \tag{12}$$

(for $a > 1$). A value of $a = 3$ works well (see parameter table in Fig. 2). The third kernel parameter is cutoff, u, defined as the number of standard deviations used in the kernel calculation. The resultant density field, which forms the basis for all steps of the CALIOP-DDA, is shown in Fig. 2(e) and (f).

Step 3 (Using Density as a Dimension: Application of the Autoadaptive Threshold Algorithm to Separate Atmospheric Layers From Background (Noise)): A key of the density-dimension algorithm is that density is considered an additional dimension of the dataset, and this additional dimension allows separation of signal areas from background areas, carried out by an autoadaptive threshold function in the geolocation-density domain. This step results in the thresholded density field, a first approximation of the signal classification [see Fig. 2(g) and (h)]. The autoadaptive capability of the algorithm implies that a threshold function is used which automatically adapts to the variable background characteristics of nighttime and daylight data, reflectance of clouds and other atmospheric layers, apparent surface reflectance, and other, including instrumental and sources of background photons.

The threshold function is controlled by the parameters: 1) t is the threshold factor; 2) T is the threshold bias offset; and 3) L is the window size parameter. The default value is $L = 2$, which results in a window size of $2L + 1 = 5$, in which case the threshold function is applied to all height-and-distance bins for five adjacent profiles and evaluated for the center profile to retain the full resolution of the data for the results.

Step 4 (Removal of Small Clusters: Determination of Cloud and Aerosol Areas (Cloud Mask)): As shown in Fig. 2(g) and (h), the threshold function leaves an approximation of signal regions that includes small clusters. After removal of small clusters (keeping clusters of minimal size m), high-density areas indicative of high-density atmospheric layer areas are mapped [see Fig. 2(i) and (j)].

Parameter Determination: The algorithm-specific parameters of the CALIOP-DDA are determined by application of sensitivity studies, which are a series of experiments during which all parameters are optimized, changing one parameter at a time and comparing to a control run. The experiments are repeated until satisfactory results are achieved.

2) Running Density Twice: As explained in Section I, a general challenge in atmospheric layer detection in Lidar data stems from the fact that data reflected from thin layers do not exceed the background intensity much, whereas optically thick layers typically result in returns that are several orders of magnitude higher.

In order to detect thin atmospheric layers in the presence of optically thick atmospheric layers, the main steps (1–4) of the DDA are carried out twice. In the first density run, a small kernel is employed to detect regions optically thick and possibly spatially narrow layers. The regions identified as thick clouds are masked out prior to the second density run. In the second density run, the *rbf* is applied with a larger kernel [larger standard deviation and anisotropy; (Fig. 2(d))], to facilitate detection of faint aerosols and clouds which typically extend for large distances. Outputs of the steps of the second density run, as illustrated in the second column in Fig. 2, indicate that this detection goal is achieved. Finally, masks from both steps are combined [see Fig. 2(k) and (l)] and a layer boundary determination algorithm is applied.

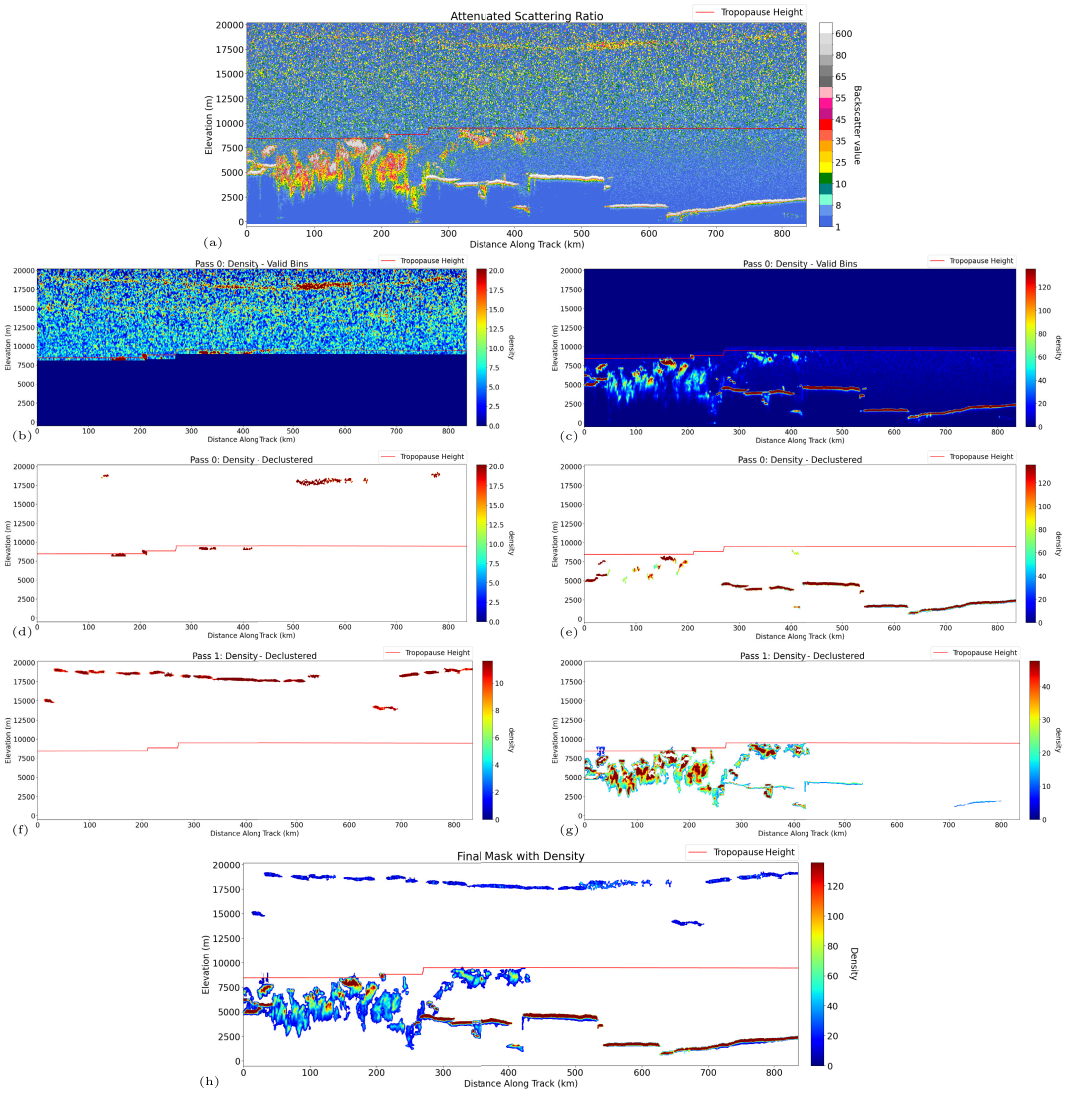
3) Determination of Layer Boundaries: Layer boundaries are determined based on the final combined mask (mask from density runs 1 and 2), using a layer amalgamation scheme that includes an upward and a downward pass to ascertain that layer have a minimal vertical extension of $layer_height_min = x1$ and minimal separation of $layer_sep_min = x2$ [42]. Resultant layer boundaries are given in Fig. 2(m).

4) Confidence: Confidence in layer detection is formulated as a measure termed "half-gap confidence," which relates the density inside an identified atmospheric layer to the density in the gap between the current layer and the layers above and below it [42], see Fig. 2(n). Confidence is calculated as $1-R$, where R is the ratio of density inside a given layer to density in the neighboring half of the gaps.

D. Tropopause Split Concept (TSC)

The Tropopause Split Concept (TSC) is a physically based mathematical concept introduced to aid in the detection of layers across large height ranges and resultant large ranges of backscatter values. Prior to running DDA steps 1–4, the dataset is split into two height regions, using the tropopause height data reported on the L1B data. Tropopause height values are obtained from the ancillary meteorological data provided by the MERRA-2 [7], [49]. The troposphere height region is defined as: $TSHR = [-0.5, T + (K + 1 + \epsilon)]$ and the stratosphere height region is defined as $SSHR = [T-(K+1+\epsilon), 20.2]$, where T is the height of the tropopause in kilometers, K is the half-height of the kernel, and ϵ is a small number of height bins. The current analysis is capped by the 20.2-km height of the second CALIOP data collection range (see Table I). The trafo function (Section IV-B) can be applied to both regions, only to *TSHR* or *SSHR*, or not at all; i.e., the input data for each height region can be TAB or ASR. CALIOP-DDA parameters can be selected for each height region separately, thus enabling a wide range of layer retrieval options (see parameter tables in the figures). After running the CALIOP-DDA (both runs) for each region, the results are combined, avoiding edge effects around the height T. To prevent edge effects, a height buffer ϵ around the tropopause height T is included in each region, adding half the maximal kernel height at the bottom of the *SSHR* and at the top of the *TSHR* before running the CALIOP-DDA

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Run Name	s	a	u	t	T	m	L	q	Q	TF
T1-top	3,6	10,40	1	0.9, 1.0	0.0001	300,600	2	0.98,0.98	0.98,0.98	true
T1-bot	3,3	10,20	1	0.9, 1.0	5.0	300,600	2	0.98,0.95	0.98,0.8	true

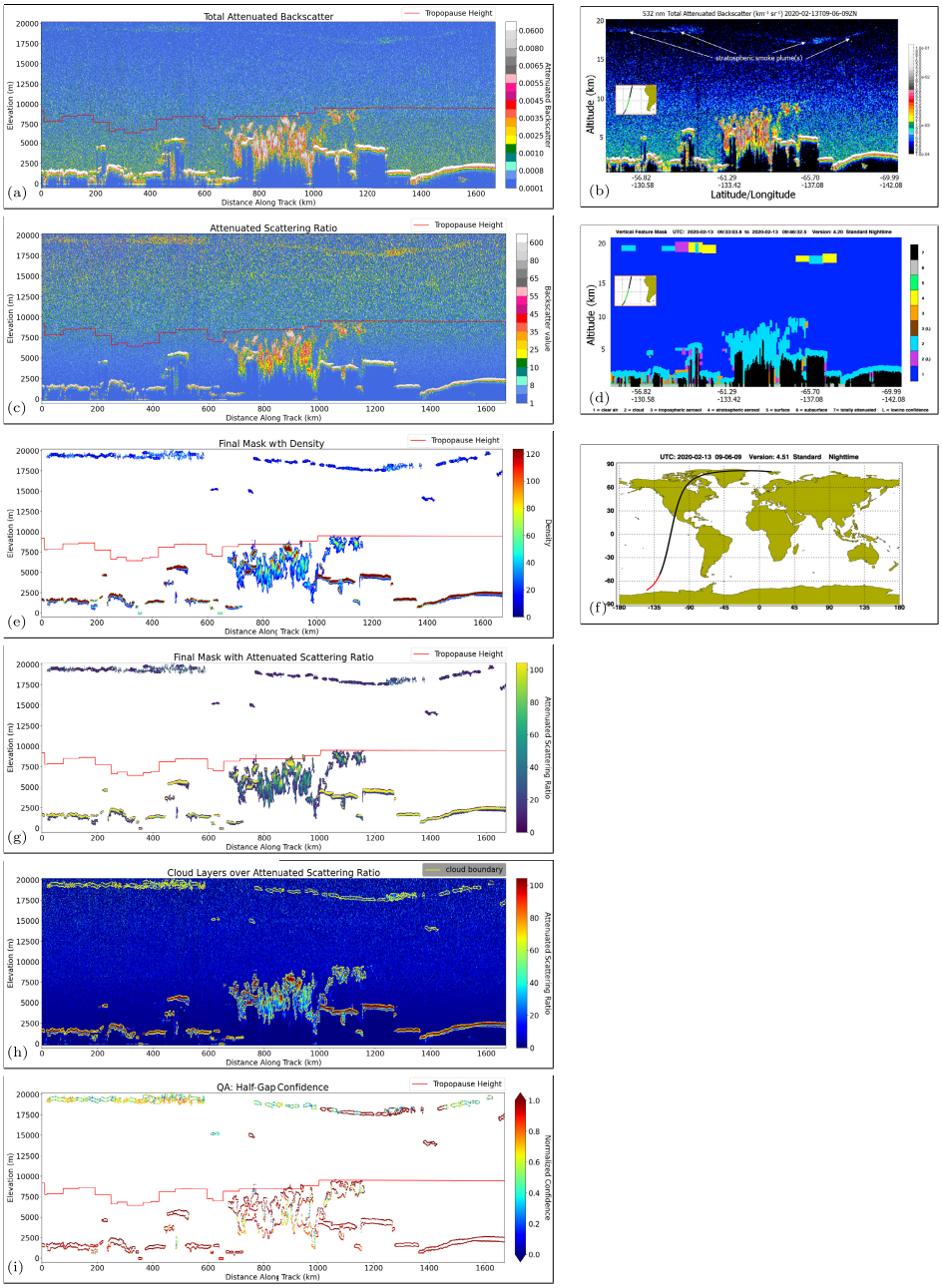
Fig. 3. Experiment T1s. Illustration of the Tropopause Split Concept, applied to detection of faint stratospheric smoke plumes using the CALIOP-DDA. Nighttime dataset. Parameter set T1. Australian wildfire examples. CALIOP data set from 2020-02-13T09-06-09. Nighttime data. Subset of data set CAL_LID_L1-Standard-V4-51.2020-02-13T09-06-09ZN (and subset of the data set used in Fig. 4). Faint aerosol layer (smoke plume) at 18000-20000 m altitude fully detected. Traces of a secondary layer are detected. (a) Full height range input data (here: Attenuated scattering ratio (ASR) data). (b) Density field, pass 1 *SSHR* (top). (c) Density field, pass 1 *TSHR* (bottom). (d) Density with decluster mask, pass 1 *SSHR* (top). (e) Density with decluster mask, pass 1 *TSHR* (bottom). (f) Density with decluster mask, pass 2 *SSHR* (top), (g) Density with decluster mask, pass 2 *TSHR* (bottom). (h) Final atmospheric layer mask, combined from density passes 1 and 2, over density pass field 1, combined for *SSHR* (top) and *TSHR* (bottom). DDA parameter descriptions as in caption of Fig. 2.

for each region. Kernel height is calculated as $2su + 1$ and half-kernel height as $su + 1$; maximal half-kernel height in bins is 10 from parameter set (T3-top) used for daytime data in density run 2 (see Section V-C). When combining the results, only the resultant values in *SSHR* and *TSHR* are used, excluding the values from the buffers. The TSC is illustrated for a subset of the stratospheric plume example (see Fig. 3).

E. Adaptation of CALIOP Data to the DDA Kernel Calculation

A goal of our work is to retain the resolution of the CALIOP original data. Since the CALIOP data have a height-dependent format, as given in Table I, with different resolutions (in both along-track and height) for different heights, we need a formalism for creating a kernel for the DDA that is valuable across the resolution boundaries. Such an algorithm is implemented as a preprocessing of the CALIOP data before the core DDA operations. Starting from the height-dependent format of the TAB data (532T TAB data), there are two ways to create a matrix: 1) average the high-resolution altitude range data or 2) increase the resolution of the low-resolution altitude range data. Both options have been implemented, as a basis for experimenting. The analysis in this article employs an algorithm for increasing the resolution for the data range between 8.2- and 20.2-km altitude, applying a smoothing operator to duplicated data from neighboring bins.

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Run Name	s	a	u	t	T	m	L	q	Q	TF
T1-top	3,6	10,40	1	0.9, 1.0	0.0001	300,600	2	0.98,0.98	0.98,0.98	true
T1-bot	3,3	10,20	1	0.9, 1.0	5.0	300,600	2	0.98,0.95	0.98,0.8	true

Fig. 4. Example of stratospheric aerosols observed 2020-02-13. Nighttime full dataset, analyzed with CALIOP-DDA using parameter set T1 and compared to CALIOP L2 results. Data set CAL_LID_L1-Standard-V4-51.2020-02-13T09-06-09ZN subset to (61.44-68.48S/ 133.52-140.10W). Left column: CALIOP-DDA results. Right Column: CALIOP data products. (a) CALIOP total attenuated backscatter (TAB) 532T data set CAL_LID_L1-Standard-V4-51.2020-02-13T09-06-09ZN, with tropopause height (red line), (b) TAB data, 15-shot average (from CALIOP L1B data), (c) CALIOP attenuated scattering ratio (ASR) data, (d) Vertical feature mask (VFM), from current (v4.51) L2 CALIOP data product, (e) Final atmospheric layer mask over density field (from density runs 1 and 2, combined for *TSHR* and *SSHR*), (f) Location map of data set used, (g) Final atmospheric layer mask over input data (ASR), (from density runs 1 and 2, combined for *TSHR* and *SSHR*), (h) Atmospheric layer boundaries from CALIOP-DDA over input data (ASR), (i) Half-gap confidence. DDA parameter descriptions as in caption of Fig. 2. Note that parameters for detection of layers in the bottom region are not optimized.

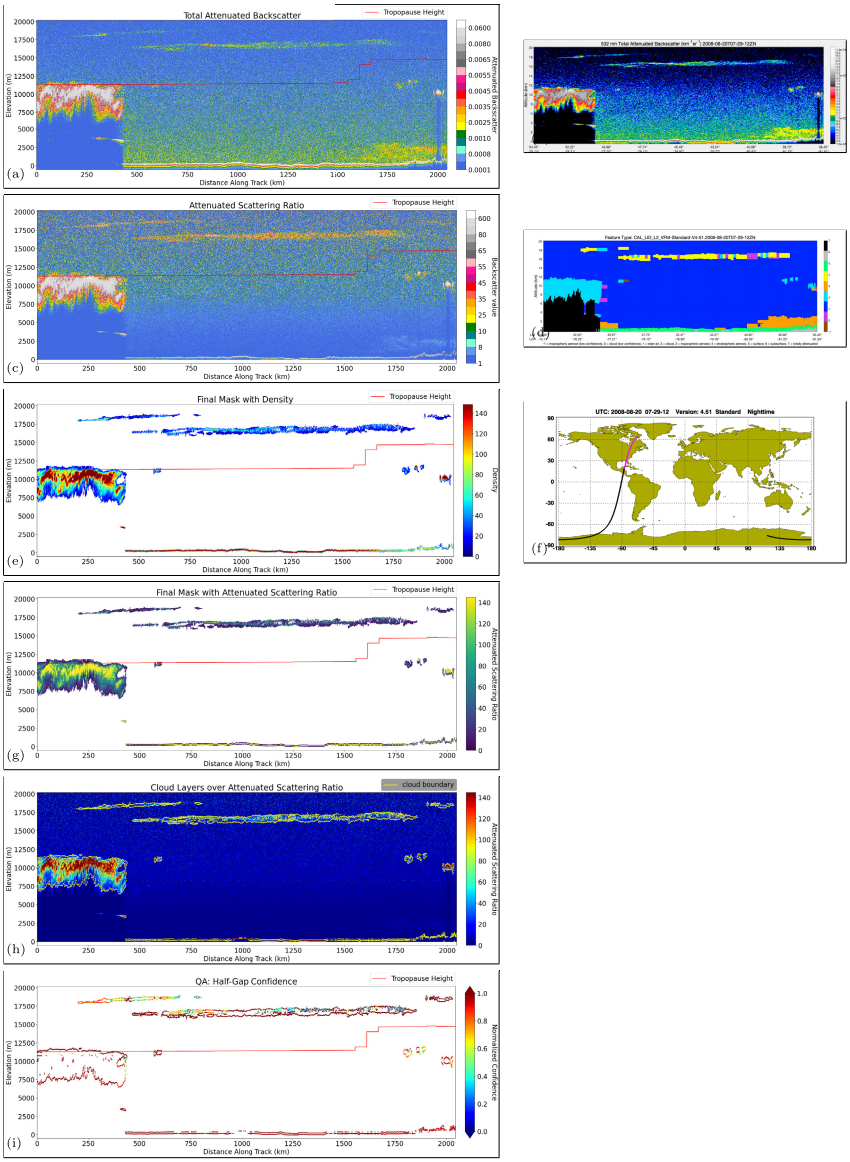
This algorithm avoids a visual break in data quality at 8.2 km and improves detection capability of layers around the 8.2-km height. We use this trick to be able to calculate the kernel of the density function (8) at full resolution, which yields better, more detailed results than if we would average data for the lower heights.

V. Application and Results

A. Detection of Faint Stratospheric Aerosols From a Smoke Plume in Nighttime CALIOP Data

As the first example, the CALIOP-DDA with TSC is applied to the dataset of a faint stratospheric smoke plume collected over the South Pacific at (61.44–68.48S/

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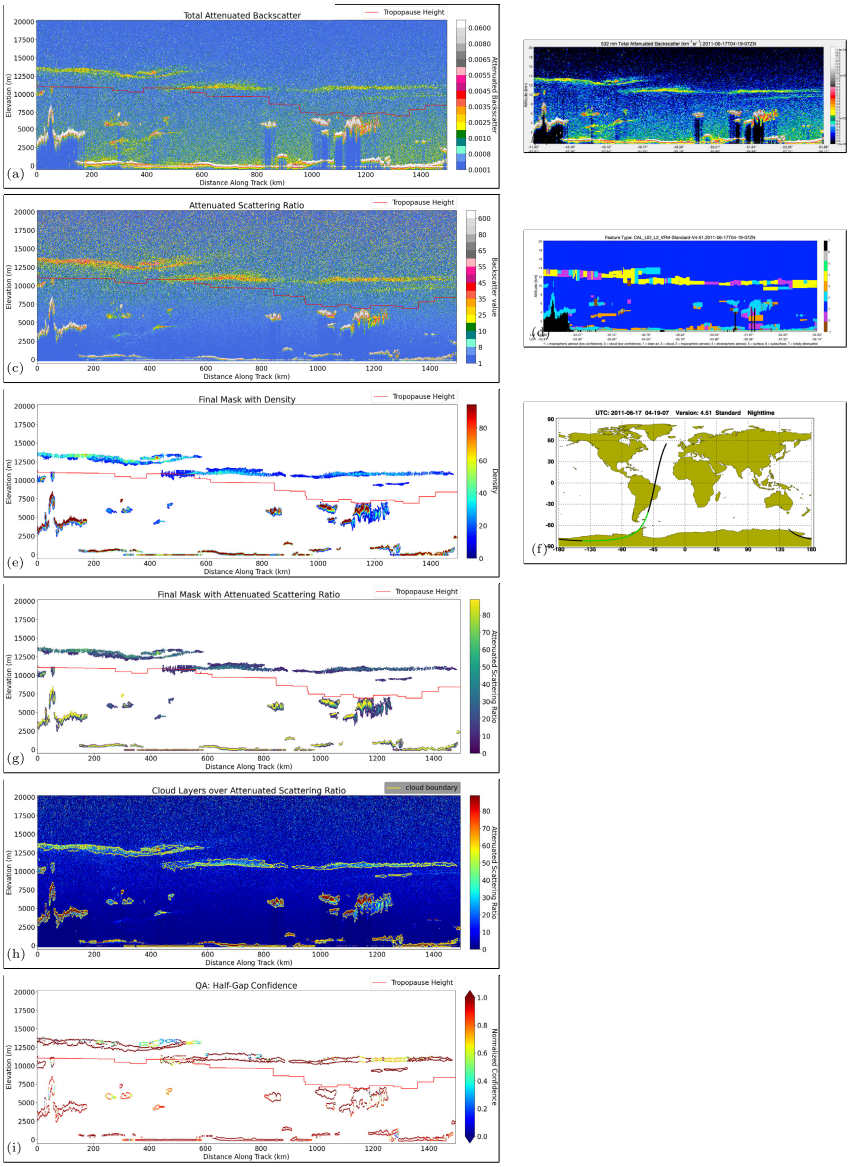
Run Name	s	a	u	t	T	m	L	q	Q	TF
T1-top	3,6	10,40	1	0.9, 1.0	0.0001	300,600	2	0.98,0.98	0.98,0.98	true
T1-bot	3,3	10,20	1	0.9, 1.0	5.0	300,600	2	0.98,0.95	0.98,0.8	true

Fig. 5. Example of stratospheric aerosols observed 2008-08-20. Nighttime full dataset, analyzed with CALIOP-DDA using parameter set T1 and compared to CALIOP L2 results. Data set CAL_LID_L1-Standard-V4-51.2008-08-20T07-29-12ZN subset to (54.45-36.40 N/ 75.14-81.93 W). Left column: CALIOP-DDA results. Right Column: CALIOP data products. (a) CALIOP TAB 532T data set CAL_LID_L1-Standard-V4-51.2008-08-20T07-29-12ZN, with tropopause height (red line), (b) TAB data, 15-shot average (from CALIOP L1B data), (c) CALIOP attenuated scattering ratio (ASR) data, (d) Vertical feature mask (VFM), from current (v4.51) L2 CALIOP data product (classes: (1) clear air, (2) cloud, (3) tropospheric aerosol, (4) stratospheric aerosol, (5) surface, (6) subsurface, (7) totally attenuated, where classes (2) and (3) have additional subclasses of low condence (2L, class 0) and (3L, class-1)) (e) Final atmospheric layer mask over density eld (from density runs 1 and 2, combined for *TSHR* and *SSHR*), (f) Location map of data set used, (g) Final atmospheric layer mask over input data (ASR), (from density runs 1 and 2, combined for *TSHR* and *SSHR*), (h) Atmospheric layer boundaries from CALIOP-DDA over input data (ASR) (i) Half-gap condence. DDA parameter descriptions as in caption of Fig. 2. Note that parameters for detection of layers in the bottom region are not optimized.

133.52–140.10W) on February 2020-02-13 used in the summary of approach (CALIOP dataset from 2020-02-13T09-06-09ZN, Subset of granule CAL_LID_L1-Standard-V4-51.2020-02-13T09-06-09ZN). The trafo function is applied to both the tropospheric height region and the stratospheric height region. To better illustrate the algorithm steps, a subset is analyzed (Experiment T1s, Fig. 3). The stratospheric aerosols are detected in the second density run. Application to the full dataset (Experiment T1f, Fig. 4) yields the full detection of the stratospheric smoke plume at 18 000–20 000 m, with a few small isolated signal areas below the main plume indicative of a secondary layer. Compared with the CALIOP VFM result [see Fig. 1(d)], the aerosol layer (smoke plume) is identified with a significantly improved horizontal extension and with a refined spatial distribution. The latter results from carrying the full resolution through the analysis steps, illustrating the difference between a data aggregation (evaluating for the center bin of each calculation of the density field, as described in Section IV-C1, step 2) and data averaging (as employed in the CALIOP VF scheme).

The algorithm specific parameters for these runs, given in the table at the bottom of Fig. 4, were determined

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Run Name	s	a	u	t	T	m	L	q	Q	TF
T1-top	3,6	10,40	1	0.9, 1.0	0.0001	300,600	2	0.98,0.98	0.98,0.98	true
T1-bot	3,3	10,20	1	0.9, 1.0	5.0	300,600	2	0.98,0.95	0.98,0.8	true

Fig. 6. Example of stratospheric aerosols observed 2011-06-17. Nighttime full dataset, analyzed with CALIOP-DDA using parameter set T1 and compared to CALIOP L2 results. Data set CAL LID L1-Standard-V4-51.2011-06-17T04-19-07ZN subset to (41.82-54.93 S/ 52.81-58.14 W). Left column: CALIOP-DDA results. Right Column: CALIOP data products. (a) CALIOP TAB 532T data set CAL LID L1-Standard-V4-51.2011-06-17T04-19-07ZN, with tropopause height (red line), (b) TAB data, 15-shot average (from CALIOP L1B data), (c) CALIOP ASR data, (d) Vertical feature mask (VFM), from current (v4.51) L2 CALIOP data product (classes as in Fig. 5(d)), (e) Final atmospheric layer mask over density eld (from density runs 1 and 2, combined for *TSHR* and *SSHR*), (f) Location map of data set used, (g) Final atmospheric layer mask over input data (ASR), (from density runs 1 and 2, combined for *TSHR* and *SSHR*), (h) Atmospheric layer boundaries from CALIOP-DDA over input data (ASR) (i) Half-gap condence. DDA parameter descriptions as in caption of Fig. 2. Note that parameters for detection of layers in the bottom region are not optimized.

using a stepwise parameter optimization. Once determined, the same parameter can be employed because of the autoadaptive capability of the threshold function. To establish this, we analyze two additional datasets with different characteristics (Section V-B).

B. Detection of Stratospheric Aerosols in Additional Nighttime Datasets: Capabilities of an Autoadaptive Algorithm

The second example captures a stratospheric aerosol, observed on August 20, 2008 in a subset of CALIOP dataset CAL_LID_L1-Standard-V4-51.2008-08-20T07-29-12ZN at (54.45–36.40N/75.14–81.93W) over northeastern North America [see Fig. 5 with map in Fig. 5(b)], extending from Middle America and the Caribbean to the east of Hudson Bay. The stratospheric layers are detected using the same parameter set (T1) that was used in the previous example (Section IV-A), for a range of backscatter values of approximately $[5 \times 10^{-4}, 5 \times 10^{-1}]$. Comparison with VFM results shows that the main stratospheric layer is largely detected in the VFM as well, whereas the secondary, higher layers are captured better by the CALIOP-DDA, and, again,

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the spatial information provided by the high-resolution DDA analysis is significantly improved compared with the VFM. Especially a very faint area of aerosol at 18 000-m altitude (located at the 2000-km along-track distance mark) is detected well, despite its occurrence above a much stronger aerosol between the ground and 3000-m altitude. The density field reflects variations in signal intensity, indicative of variations in optical depth, within the stratospheric layers [see Fig. 5(e) and (g)]. Confidence is high (0.8–1.0) to good (0.4–0.7) for all stratospheric layer retrievals [see Fig. 5(i)] and no false positives occur.

A tropospheric cloud layer of complex density and shape is well-detected at 7000–12 000-m altitude, it has much higher density than the stratospheric layer above it [see Fig. 5(e) and (g)–(i)]. Of note, while this cloud layer crosses the tropopause, its extension and boundaries are captured as those of a single feature (undisturbed by the tropopause split), which demonstrates that the split-and-recombine modules of the algorithm function correctly [see Fig. 5(e) and (g)–(i)]. The parameters for the troposphere, however, are not fully optimized and a low layer classified as tropospheric aerosol in the VFM (near 36.40N/−81.61E) is missed [see Fig. 5(d) compared with Fig. 5(e), (g), and (h)], which may be attributed to the use of ASR data as input for the *TSHR*. Ground is detected by the DDA, albeit not classified as such in the current algorithm version [see Fig. 5(e) and (g)–(i)].

The third example stems from data collected over Antarctica and the South Pacific [see Fig. 6(a)–(c) and (f)] on June 17, 2011 (CALIOP dataset CAL_LID_L1-Standard-V4-51.2011-06-17T04-19-07ZN [subset (41.82–54.93S/52.81–58.14W)]. Again, the CALIOP-DDA is applied with the same parameter set T1 and ASRs as input data (see Fig. 6). The stratospheric layers are fully retrieved at full resolution. The density field [see Fig. 6(e)] shows that the density or potentially the material of the layer has significant internal variability, which may warrant identification of two different layers (nearly 600–800-km along-track). Confidence for detection of the more dense regions is higher than for the more faint layers, while still good overall [see Fig. 6(i)]. Compared with the layer identification and classification in the VFM [see Fig. 6(d)], the stratospheric layers detected by the CALIOP-DDA show their characteristic high-resolution detail [see Fig. 6(e), (g), and (h)], whereas the VFM results in a single layer that is horizontally connected across the entire analysis range. Classification in the VFM switches frequently between stratospheric aerosol (class 4, correct), cloud (classes 2 and 2L, incorrect), and tropospheric aerosol (class 3L, incorrect).

Examination of the *TSHR* shows that almost all layers are detected that are visually apparent in the TAB and the ASR plots (see Fig. 6(a) and (c), respectively), except a faint layer at 300–400-km along-track is only partly detected (possibly a tropospheric aerosol). On the other hand, the VFM [see Fig. 6(d)] identifies tropospheric aerosols—incorrectly—also at (−51.67N/−56.58E), in a region where the signal is fully attenuated by thick clouds above. This situation is different from the missed faint layers at 300-400 km along-track as well as that in example 2 (analyzed above, Fig. 5). As before, this is a result of a lack of optimal parameter determination for the *TSHR*.

In summary, the analysis in this section suggests that the parameter set T1 is suitable for the detection of faint stratospheric layers in nighttime data.

C. CALIOP-DDA-Analysis of Daytime Data

To demonstrate the functionality of the CALIOP-DDA for daytime data, we analyze two examples, 1) a dataset of another Australian wildfire smoke plume in the stratosphere, collected on Janauary 19, 2020 (dataset CAL_LID_L1-Standard-V4-51.2020-01-19T02-14-38ZD) over the southern Pacific east of Australia at (50.02–28.97 S/179.31 W–173.92 E) (see Fig. 7) and 2) the CPL validation set (see Fig. 8, Section VI). The challenge of layer detection and especially of faint layer detection is much larger for daytime data, because the background is much higher than for nighttime data, as illustrated in the example of CALIOP daytime data collected on January 19, 2020. In addition, large along-track variations in background intensity exist, further obscuring the atmospheric layer structure [see Fig. 7(a) and (b)]. To meet this retrieval challenge, we use a much larger kernel for analysis of daytime data with the CALIOP-DDA (see parameter table T3 at the bottom of Fig. 7). Aggregation over larger regions, especially in the horizontal direction, reduces the effect of the along-track variations and of the increased background. In addition, the threshold function is applied for a larger window (window size parameter $L = 18$, resulting in a window size of $2L + 1 = 37$ profiles), whereas for nighttime data, a window of five profiles (window size $L = 2$) is used.

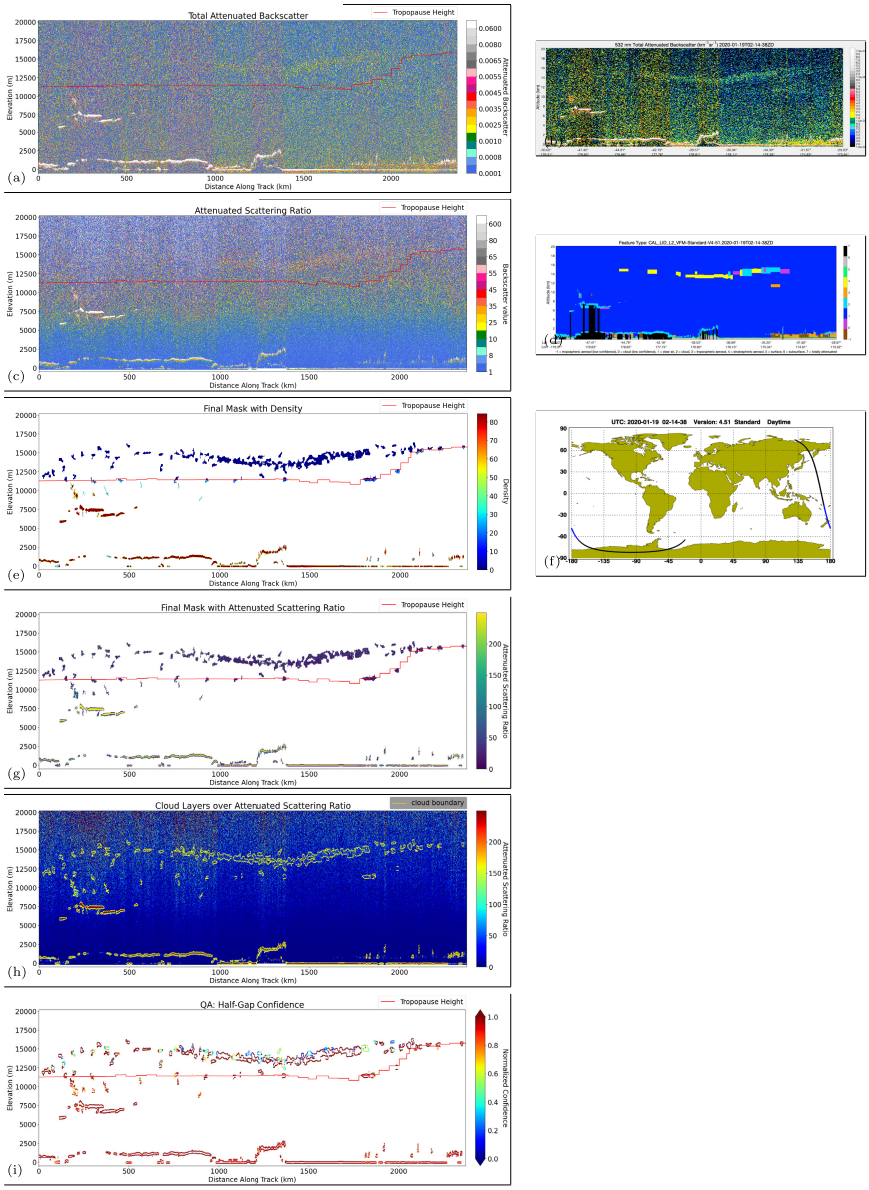
As a result, the smoke plume is detected [see Fig. 7(e), (g) and (h)] with greater extension than in the VFM [see Fig. 7(d)]. The VFM misses the regions where the stratospheric smoke plume is faint (left part of the wave). Other than for nighttime data, the trafo function is not applied above the tropopause, because the ASR is poorly affected by the characteristics of daytime data (compare Fig. 7(a) (TAB) and 7(c) (ASR). In this example, all atmospheric layers (clouds) that are recorded in the *TSHR* are also captured by the CALIOP-DDA. In summary, the CALIOP-DDA facilitates detection of stratospheric aerosols also in CALIOP daytime data.

VI. Validation of Faint-Layer Retrieval With the CALIOP-DDA Using CPL Data and Comparison With Results From the Operational CALIOP Algorithm

A. Validation Using CPL Data

To evaluate the performance of the CALIOP-DDA, we utilize CPL data collected during an airborne CALIPSO validation campaign that was conducted in July 2009, a year after the launch of the CALIPSO satellite. CPL is a backscatter Lidar designed to operate simultaneously at three wavelengths: 1064, 532, and 355 nm. The purpose of the CPL is to provide multiwavelength measurements of cirrus, subvisual cirrus, and aerosols with high temporal and spatial resolution (https://airbornescience.nasa.gov/instrument/Cloud_Physics_Lidar, date accessed on June 11, 2024). The

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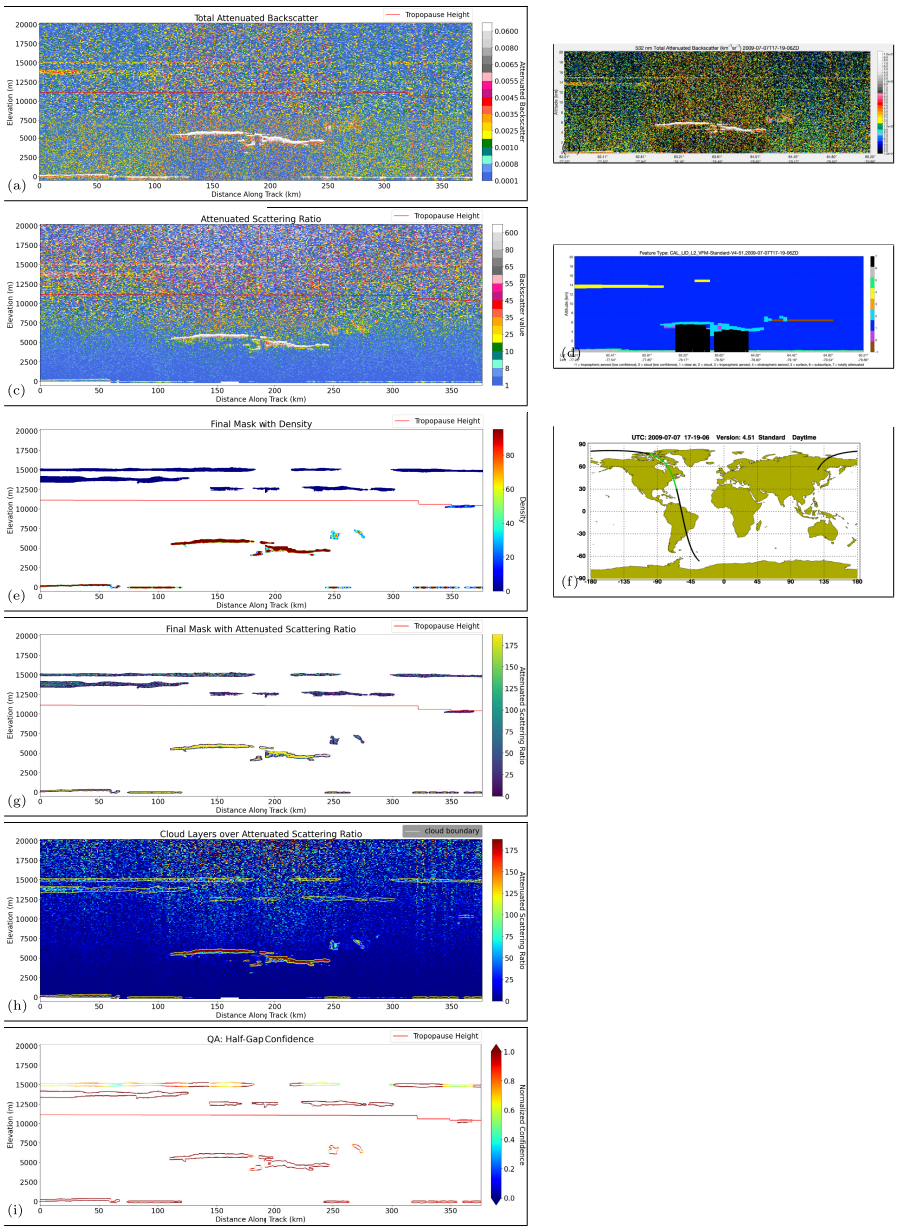
Run Name	s	a	u	t	T	m	L	q	Q	TF
T3-top	6,9	40,80	1	0.9, 1.0	0.0002	600,600	18	0.94,0.92	0.98,0.98	false
T3-bot	3,3	10,20	1	0.9, 1.0	5.0	300,300	18	0.97,0.95	0.98,0.8	true

Fig. 7. Example of application of the CALIOP-DDA to analysis of CALIOP daytime data of stratospheric aerosols observed 2020-01-19. Daytime full dataset, analyzed with CALIOP-DDA using parameter set T3 and compared to CALIOP L2 results. Data set CAL_LID_L1-Standard-V4-51.2020-01-19T02-14-38ZD subset to (50.02 - 28.97 S/ 179.31 W - 173.92 E). Left column: CALIOP-DDA results. Right Column: CALIOP data products. (a) CALIOP TAB 532T data set CAL LID L1-Standard-V4-51.2020-01-19T02-14-38ZD, with tropopause height (red line), used in analysis of the *SSHR* (top), (b) TAB data, 15-shot average (from CALIOP L1B data), (c) CALIOP ASR data, used in analysis of the *TSHR* (bottom), (d) Vertical feature mask (VFM), from current (v4.51) L2 CALIOP data product (classes as in Fig. 5(d)), (e) Final atmospheric layer mask over density field (from density runs 1 and 2, combined for *TSHR* and *SSHR*), (f) Location map of data set used, (g) Final atmospheric layer mask over ASR data, (from density runs 1 and 2, combined for *TSHR* and *SSHR*), (h) Atmospheric layer boundaries from CALIOP-DDA over ASR data, (i) Half-gap condence. DDA parameter descriptions as in caption of Fig. 2.

CPL utilizes state-of-the-art technology with a high repetition rate, low-pulse energy laser, and photon-counting detection. Vertical resolution of the CPL measurements is fixed at 30 m; horizontal resolution can vary but is typically about 200 m. The CPL measures range-resolved profiles of volume 180° backscatter coefficients. The CPL was operated on the NASA ER-2, a research aircraft that typically flies at about 20-km altitude. In consequence, the CPL records most (94%) of the Earth's atmosphere; however, the signal quality close to the aircraft (just below 20 km) is poor. The thin bright line seen at 20 km in Fig. 9(a) is not an atmospheric feature, but is instead an artifact produced by the CPL data acquisition system.

During the July 2009 validation campaign, the ER-2 underflew CALIPSO (see Fig. 8) In our comparison, we use CPL 532-nm attenuated backscatter [see Fig. 9(a)] for a region where a stratospheric aerosol layer, resultant from a distant volcanic eruption, was recorded on July 7, 2009. The time of exact co-occurrence of CPL and CALIOP data collection is at 17:55:25 UTC at a location at approximately 63.5N

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Run Name	s	a	u	t	T	m	L	q	Q	TF
T3-top	6,9	40,80	1	0.9, 1.0	0.0002	600,600	18	0.94,0.92	0.98,0.98	false
T3-bot	3,3	10,20	1	0.9, 1.0	5.0	300,300	18	0.97,0.95	0.98,0.8	true

Fig. 8. Analysis of a CALIOP daytime data set for validation of faint layer retrieval with the CALIOP-DDA using CPL data. Daytime full dataset, analyzed with CALIOP-DDA using parameter set T3 and compared to CALIOP L2 results. Data set CAL_LID_L1-Standard-V4-51.2009-07-07T17-19-06ZD subset to (62.01-65.21 N/ 77.23-79.89 W). Left column: CALIOP-DDA results. Right Column: CALIOP data products. (a) CALIOP TAB 532T data set CAL_LID_L1-Standard-V4-51.2009-07-07T17-19-06ZD, with tropopause height (red line), used in analysis of the *SSHR* (top), (b) TAB data, 3-shot average (from CALIOP L1B data), (c) CALIOP ASR data, used in analysis of the *TSHR* (bottom), (d) Vertical feature mask (VFM), from current (v4.51) L2 CALIOP data product (classes as in Fig. 5(d)), (e) Final atmospheric layer mask over density field (from density runs 1 and 2, combined for *TSHR* and *SSHR*), (f) Location map of data set used, (g) Final atmospheric layer mask over ASR data, (from density runs 1 and 2, combined for *TSHR* and *SSHR*), (h) Atmospheric layer boundaries from CALIOP-DDA over ASR data, (i) Half-gap condence. DDA parameter descriptions as in caption of Fig. 2.

east of Hudson Bay. Data are analyzed for (62.01–65.21N/ 77.23–79.89W).

The CALIOP-DDA was applied to the matching CALIPSO dataset (CAL_LID_L1-Standard-V4-51.2009-07-07T17-19-06ZD), and results are shown in Fig. 9. Two faint stratospheric aerosol layers, which are almost horizontal, are detected and measured by the DDA. In comparison, the CALIOP VFM retrieves only one layer and only partially, but classifies it correctly as stratospheric aerosol. The DDA is applied using the same daytime parameter set T3 (see table at the bottom of Fig. 8) as in the daytime example in Section IV-C, to TAB for the *SSHR* (no trafo) and to ASR data (trafo true) for the *TSHR*. The cloud layer in the troposphere is also captured well [see Fig. 8(e) and (g)–(i)].

To compare with the CPL results, we inspect: 1) CPL layer heights as reported on the CPL data product [see Fig. 9(a)] and 2) CALIOP-DDA layer heights plotted over CPL attenuated backscatter data (normalized radiometric backscatter

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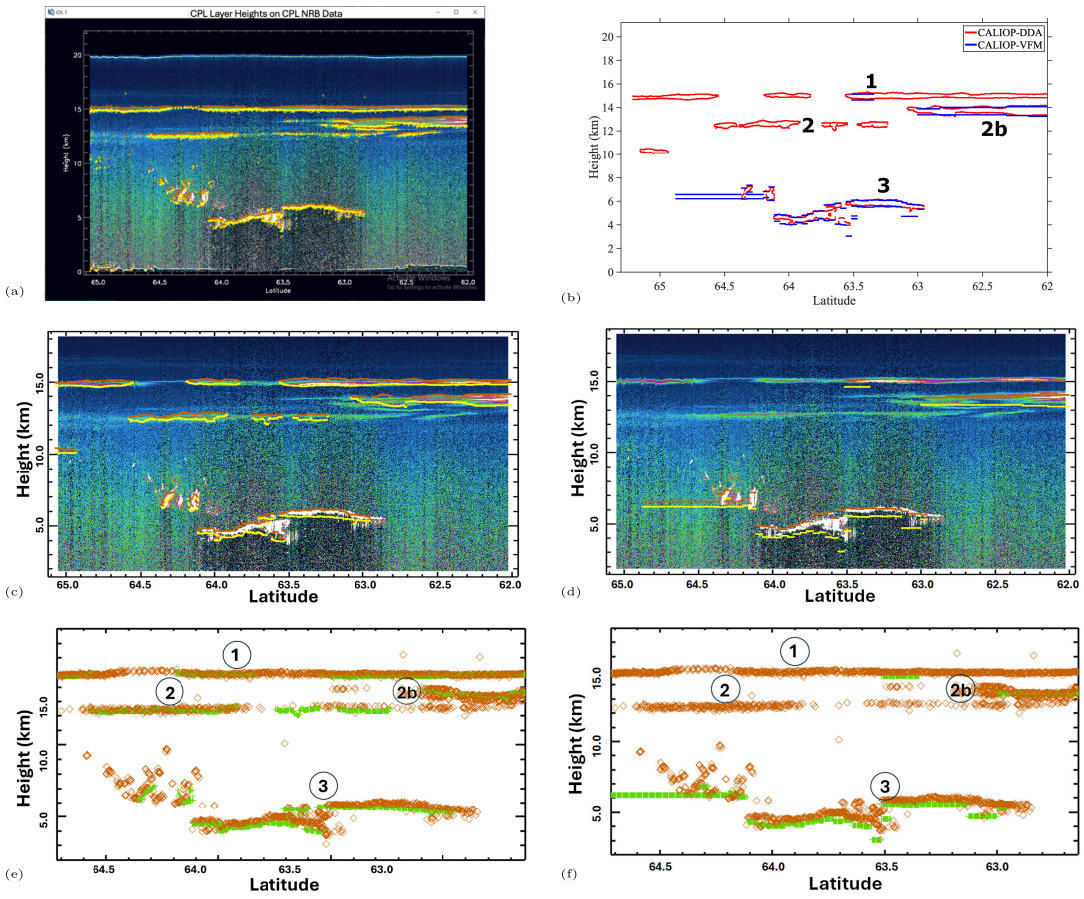


Fig. 9. Evaluation of CALIOP-DDA results using CPL data and comparison with CALIPSO VFM results. Comparison of retrieval of stratospheric aerosols from CALIOP data using the CALIOP-DDA with layer heights from CPL data, based on CPL underights of CALIPSO. 2009-07-07. Region is Canada near Hudson Bay. (a) CPL data with layer boundaries from CPL data analysis. red - layer top, yellow - layer bottom. (b) Comparison of layer boundaries from CALIOP-DDA (red) and CALIOP-VFM (green). (c) CPL data with layer boundaries from CALIOP data analysis, resultant from CALIOP-DDA with tropopause split. Parameter set T3. red - layer top, yellow - layer bottom. (d) CPL data with layer boundaries from CALIOP-VFM 4.51. red - layer top, yellow - layer bottom. (e) Comparison of CPL layer heights (red) and CALIOP-DDA layer heights (green) used in calculation of differences (Table II). (f) Comparison of CPL layer heights (red) and CALIOP-VFM 4.51 layer heights (green) used in calculation of differences (Table II).

(NRB) data) [see Fig. 9(c)]. As a result, the stratospheric aerosol is captured by the DDA to the extent it is measured by the CALIOP sensor, and no false positives are reported. The height of the top and bottom layer boundaries is reported correctly. However, comparison with the CPL NRB shows that the airborne CPL records up to three or four stratospheric aerosol layers, whereas the CALIPSO sensor only records maximally two in the same profiles, but three in total [labeled (1), (2), and (2b) in Fig. 9(e) and (f)]. In summary, the evaluation of the CALIOP-DDA supports the finding that the algorithm facilitates the detection of faint stratospheric aerosols, even in daytime conditions.

The comparison of profile heights [see Fig. 9(c)] for tropospheric clouds is affected by the fact that the height determination in the CPL data looses accuracy with increasing distance from the aircraft. In general, the coincidence of tropospheric heights and their layer boundaries between CALIOP-DDA and CPL is good, indicating that the CALIOP-DDA also works correctly for layer detection in the troposphere, especially for cloud detection.

Numerical comparison of the CPL layer top heights, for the three main layers identified by both the CPL and the CALIOP-DDA detection scheme [see Fig. 9(d), DDA CALIOP layer heights (green) with the CPL heights (red)], is summarized in Table II. Differences in layer heights, calculated as [CPL-Layer-Top-Height minus CALIOP-DDA-Layer-Top-Height] and [CPL-Layer-Bottom-Height minus CALIOP-DDA-Layer-Bottom-Height], show a very good agreement for the top two layers, which are stratospheric aerosols from a volcanic eruption. For all three layers, the average difference of the top layer boundaries is less than the resolution of a single height bin (30 m), whereas the difference for the bottom boundaries is around two bins in size for the stratospheric layers and three bins for the tropospheric layer. Minima and maxima of the differences across the analysis data range are low for the two stratospheric layers (L1, L2), reaching ten bins, especially considering that the CPL tracks and the CALIPSO tracks diverge with distance from the point of exact coincidence. A visual interpretation of the comparison figure highlights that the CALIOP-DDA is able to pick up the central section of layer L2 better than the CPL data analysis [green symbols at approximately 15-km height near 63°–63.5°

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TABLE II

DIFFERENCE IN LAYER HEIGHTS BETWEEN CPL AND CALIOP DETECTION SCHEMES (CALIOP-DDA AND CALIOP-VFM) FOR THE JULY 7, 2009 CROSSOVER SEGMENT. FOR EACH COLUMN, THE DIFFERENCE IN HEIGHT FOR THE LAYER TOP IS LISTED FIRST, FOLLOWED BY THE HEIGHT DIFFERENCE FOR THE LAYER BOTTOM. STATISTICS FOR $Q = 0.9$ ACCOUNT FOR OUTLIERS BY USING ONLY HEIGHT DIFFERENCES BELOW THE 0.9-QUANTILE

	Layer 1 (top, bottom)	**Layer 2 (top, bottom)**	**Layer 2b (top, bottom)**	**Layer 3 (top, bottom)**
CPL layer height minus CALIOP-DDA layer height				
Average (m)	29.0, 63.2	13.1, 66.3	-34.3, -41.3	-18.0, 107.1
Minimum (m)	-170.0, -290.0	-289.0, -199.0	-470.0, -679.0	-1783.0, -2023.0
Maximum (m)	219.0, 220.0	221.0, 311.0	250.0, 340.0	1576.0, 1666.0
Average, Q=0.9 (m)	29.0, 62.1	7.6, 61.3	-12.2, -42.6	-9.9, 108.6
Minimum, Q=0.9 (m)	-111.0, -260.0	-169.0, -169.0	-410.0, -529.0	-1603.0, -1813.0
Maximum, Q=0.9 (m)	189.0, 190.0	191.0, 251.0	220.0, 280.0	1366.0, 1426.0
Unique Height Estimates	105	82	65	219
CPL layer height minus CALIOP-VFM layer height				
Average (m)	81.7, 256.3	———	32.8, 73.8	-112.1, 555.3
Minimum (m)	-60.2, 28.4	———	-424.4, -726.3	-1873.3, -1874.3
Maximum (m)	179.8, 328.4	———	205.0, 502.7	1331.9, 2349.4
Average, Q=0.9 (m)	81.5, 249.8	———	37.4, 69.0	-120.3, 550.0
Minimum, Q=0.9 (m)	-0.2, -28.4	———	-365.0, -336.3	-1663.5, -1634.4
Maximum, Q=0.9 (m)	149.8, 268.4	———	175.0, 443.7	762.9, 2109.
Unique Height Estimates	1	0	3	52

latitude, as shown in Fig. 9(e)]. For the tropospheric layer, L3, the comparison values have a much larger spread, mainly due to multiple scattering in the CALIOP data. Analysis of CPL data from the validation campaign has shown that when optical depths get large, multiple scattering allows CALIOP to penetrate much deeper into the cloud than CPL.

B. Comparison of CALIOP-DDA Results With CALIOP-VFM Results Using CPL Data

A visual comparison of faint stratospheric layer detection capabilities of the CALIOP-DDA with the detection results reported in the CALIPSO VFM (v4-51) is given in the right column of each of Figs. 4–8. Advances in layer detection include: 1) detection of extremely faint layers that are currently missed or partly missed and 2) an increase in resolution of the layer detection, which leads to a retrieval of details in layer shape and spatial distribution. Here, we utilize the CPL data to examine the layer retrieval reported on the CALIPSO L2 data product in the VFM [see Fig. 9(d) and (f)] and compare it to the results from the CALIOP-DDA [see Fig. 9(c) and (e)]. A comparison of layer boundaries from the CALIOP-DDA and the CALIOP-VFM [see Fig. 9(b)] illustrates that the CALIOP-DDA retrieves three different stratospheric layers (layers 1, 2, and 2b), whereas the VFM reports only short sections of layers 1 and 2b and misses layer 2 entirely. Both algorithms retrieve the tropospheric layer (layer 3) between latitudes 63.0 and 64.1, with more spatial detail reported by the CALIOP-DDA. For a quantitative comparison, results are summarized in Table II. The most stringent difference between the CALIOP-DDA and the CALIOP-VFM lies in the substantial increase of unique layer heights reported as results, reflecting both the increased layer detection capability of the CALIOP-DDA and the significant increase in spatial resolution of layer boundaries. Differences in layer heights between CPL and CALIOP-VFM (calculated as [CPL-Layer-Top-Height minus CALIOP-VFM-Layer-Top-Height] and [CPL-Layer-Bottom-Height minus CALIOP-VFM-Layer-Bottom-Height] are also larger than for the equivalent height differences between CPL and CALIOP-DDA, for example, an average difference of (29.0, 63.2 m) for CPL-minus-CALIOP-DDA heights for layer 1 corresponds to an average difference of (81.7, 256.3 m) for CPL-minus-CALIOP-VFM heights. For layer 2b, the comparison results are much closer, with an average difference of (−34.3, −41.3 m) for CPL-minus-CALIOP-DDA heights and (32.8, 73.8 m) for CPL-minus-CALIOP-VFM heights, noticing that the height resolution of the DDA results is 30 m and that of the VFM is 60 m. In summary, the quantitative comparison confirms and details the conclusion that stratospheric layer detection using the CALIOP-DDA presents a significant advance over the results from the CALIOP-VFM.

VII. DISCUSSION

A. Signal–Noise Separation

Application of the CALIOP-DDA to daytime data does not require prior denoising of the CALIOP data. The DDA includes an efficient signal–noise separation algorithm module, which works for daytime and nighttime data, this is the auto-adaptive threshold function described in step 3 in Section IV-C1. The differences in signal to background are handled by using different algorithm-specific parameter sets for nighttime and daytime data. For nighttime data in the stratospheric height region, the CALIOP-DDA is applied to 532T ASR data, whereas for daytime analysis, the DDA

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is applied to 532T TAB data. An advantage of integrating the signal–noise separation into the algorithm flow is that information inherent in the CALIOP data is not reduced early on in the analysis, as can be the case for some denoising methods (see [58], [59]). The use of an auto-adaptive threshold function, applied with parameters specific to daytime conditions, is key to the capability of the DDA to detect extremely faint layers, even in the presence of much stronger layers in the same profiles. Use of a data aggregation, rather than a data averaging approach, allows the CALIOP-DDA to retain high resolution of the output of layers with generally high confidence (see Figs. 6 and 7). Evaluation using CPL data supports these findings. Furthermore, the evaluation with CPL data actually demonstrates that the DDA handles noise during daytime rather well, as not just one but two layers are detected, whereas the current CALIOP algorithm only finds at most one layer per location/profile (Sections V-C and VI and Figs. 7 and 8) The need to use a larger kernel for daytime than for nighttime data analysis does not generally lead to significant loss of detail, because faint stratospheric smoke layers typically have a very large horizontal extension.

Application of neural networks for signal–noise separation has been explored [60]; however, the generalizability of these methods has yet to be demonstrated. Especially, convolutional neural networks (CNNs) tend to not generalize well or require extremely large datasets for training [61], [62].

B. Computational Efficiency

The DDA is a conceptually complex but computationally simple and efficient algorithm. The computationally most expensive step is the calculation of the density field. As the mathematical description of the density calculation (step 2, Section IV-C1) shows, this requires mostly linear algebra, which is inexpensive. The DDA-atmos is the operational algorithm for ICESat-2 atmospheric layer detection, reported on ICESat-2 ATLAS data product ATL09 [42], [44] and has proven to be computationally efficient and not very expensive. The CALIPSO data have similar (a bit lower) resolution than ICESat-2 ATLAS data, CALIPSO has three channels (532T, 532S, and 1064), whereas ICESat-2 has a single 532-nm channel, but records three strong beams which are analyzed for the atmospheric data products [43], suggesting that computational cost may be similar. Thus, the use of the CALIOP-DDA as an operational algorithm for new CALIOP-L2 data appears to be feasible. We will be running tests in an operational environment to ascertain this in the near-term future.

The parameter optimization has to be carried out only once to determine optimal settings of the algorithm-specific parameters. After the algorithm-specific parameters have been determined, the DDA runs autoadaptively without further optimization.

VIII. Conclusion

The objective of this article is to demonstrate a new capability to detect faint stratospheric aerosols in atmospheric Lidar data from NASA's CALIPSO Mission (April 28, 2006–August 1, 2023 with science data available 12 June 2006–30 June 2023), facilitated by the development of a new algorithm, the CALIOP-DDA.

The work in this article is motivated by the following problem. Faint aerosols from wildfires, distant volcanic eruptions, and Asian and Saharan dust storms play important roles in the Earth's atmosphere system, as they provide essential information on air quality and pollution, airline safety, climate radiative forcing, monitoring of fires and volcanic eruptions, environmental safety, and weather forecasting. Observations of atmospheric layers are collected using satellite Lidar measurements, which provide information on the vertical structure of the atmosphere. Yet tenuous layers often remain undetected because they tend to be characterized by a low signal gradient to their surroundings and poor signal-to-noise ratios. Such layers are typically diffuse remnants of volcanic eruptions and pyro-cumulonimbus smoke injections in the lower stratosphere and Asian dust, volcanic plumes, and subvisible cirrus clouds in the upper troposphere. Tenuous aerosols are recorded in the CALIOP data from NASA's CALIPSO mission, but especially stratospheric aerosol layers often escape detection and classification partly or entirely in the current CALIOP data analysis scheme. An operational optical thickness threshold of 0.01 has been established for CALIPSO data [28], [29], which is much lower than the threshold for subvisible clouds.

To solve the problem of detection and height measurement of tenuous atmospheric layers, we introduce the CALIOP-DDA. The CALIOP-DDA is based on the Density-Dimension Algorithm for atmospheric data collected with the ATLAS of NASA's ICESat-2 mission [41], [42], the DDA-atmos, which is the operational algorithm for atmospheric layer detection as reported on ICESat-2 ATLAS atmospheric data product ATL06 [43], [44].

The DDA builds on concepts from artificial intelligence and spatial statistics. The center of the DDA is the calculation of the density field, on which all other algorithm steps are based. An autoadaptive threshold function yields signal–background separation, using the density field as data in an additional dimension. Removal of small clusters leads to the generation of cloud and aerosol areas, identified by a mask. The detection of faint layers is facilitated by the concept of running density twice, with different parameters that control the kernel of the algorithm. The masks from both density runs are combined into a final mask, and finally, layer boundary heights are derived from this final mask.

The CALIOP-DDA utilizes CALIOP level L1B data TAB data (532T TAB data, v4-51) as input into its analysis, signal classification and layer detection scheme [49]. Alternatively, the CALIOP-DDA can use ASR data, obtained from the TAB data by a transformation described here, following [35]. Prior to the core DDA steps, the CALIOP data are prepared for input into a matrix by preprocessing steps that result in datasets of the full resolution of the 0–8.2-km height range (334-m along-track and 30-m height bins), extended to 20-km altitude. This resolution is retained in the DDA output. The analysis in this study focuses on the optimization and application of the CALIOP-DDA for the detection of faint stratospheric aerosols,

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which is aided by the TSC using ancillary data on tropopause height reported on the CALIOP L1B data products.

The CALIOP-DDA results in: 1) a signal–background separation (classification of signals); 2) detection of atmospheric layers; 3) height measurements of atmospheric layers in the form of layer boundary heights (top and bottom); and 4) a confidence measure. The CALIOP-DDA also provides density fields.

Here, the CALIOP-DDA is applied to several nighttime and daytime datasets, which have recorded returns from stratospheric aerosols. The DDA is controlled by a set of algorithm-specific parameters. Two different sets of algorithm-specific parameters, one for nighttime data analysis and one for daytime data analysis, are utilized, following a parameter optimization. Data aggregation over larger regions, especially in the horizontal direction, reduces the effect of the along-track variations and of the increased background typical of daytime data. Once the set of algorithm-specific parameters is determined, the DDA runs auto-adaptively (i.e., without further changes of parameters). The results obtained from the data analyzed here support this.

Results: The main result is that the CALIOP-DDA facilitates detection of faint stratospheric aerosols at high resolution. The CALIOP-DDA allows detection of extremely tenuous layers in the presence of complex types of other cloud and aerosol layers across a large range of optical thicknesses, without creating false positives. The DDA allows detection of stratospheric aerosols that have a backscatter intensity that is up to three orders of magnitude lower than that of cloud layers in the troposphere, which requires a detection capability of really faint layers in the presence of optically thicker layers. However, the CALIOP-DDA is also able to detect different layer types that are closer in backscatter intensity (ranging, for example, across a backscatter intensity factor of one or two orders of magnitude).

Comparison of the CALIOP-DDA results to detection with the detection and classification results of the existing (official) algorithms (v4-51), as reported in the VFM under [49], [54] supports the following findings.

1) Layers detected with the CALIOP-DDA include faint aerosols that escaped detection with the current (official) CALIOP (v4-51) scheme or were only partly detected. These layers are classified further as either tropospheric or stratospheric based on their altitudes with respect to the tropopause height.
2) Stratospheric layers detected with the current CALIOP algorithms are also detected with the CALIOP-DDA.
3) A major advance in layer detection afforded by the DDA lies in the high-resolution representation of all layers (i.e., all types of layers and at all heights). This result is attributed to the use of a data aggregation method, rather than an averaging method, in the calculation of the density field, which is retained in all following DDA steps. In consequence the DDA provides results at the original 334-m along-track/30-m height resolution of the 0–8.2-km CALIOP data, with the same resolution extending to 20.2-km altitude due to the data upsampling scheme, which is necessary for kernel operation across height boundaries.

Thus, the results indicate that the CALIOP-DDA furthers the four science goals of the CALIPSO Mission, especially the goal of detection of clouds and aerosols at the highest possible spatial resolution [17].

To ascertain the generalizability and robustness of the algorithm, the concepts were evaluated for a number of different datasets, selected to represent different types of cloud and aerosol scenarios. Approximately 30 different CALIOP datasets were analyzed. Most tropospheric layers detected by the official CALIOP algorithms are also detected by the CALIOP-DDA; however, in a few locations, tropospheric layers identified as in the CALIOP VFM as aerosols are missed by the CALIOP-DDA with the currently used parameter set, likely because the parameters are not fully optimized for tropospheric aerosol detection.

DDA analysis results are evaluated using airborne data collected with the CPL during the July 2009 CALIPSO validation campaign, based on a CPL dataset that recorded stratospheric aerosols from a distant volcanic eruption. The stratospheric aerosol is captured by the DDA to the extent it is measured by the CALIOP sensor, and no false positives are reported. Numerical analysis of height differences of CPL layer heights, from the standard CPL analysis scheme, and CALIOP-DDA-derived layer heights, shows very good agreement, with an average difference of less than a single height bin (30 m).

In conclusion, the CALIOP-DDA holds promise as the algorithmic basis for a future improved, high-resolution CALIPSO data product.

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